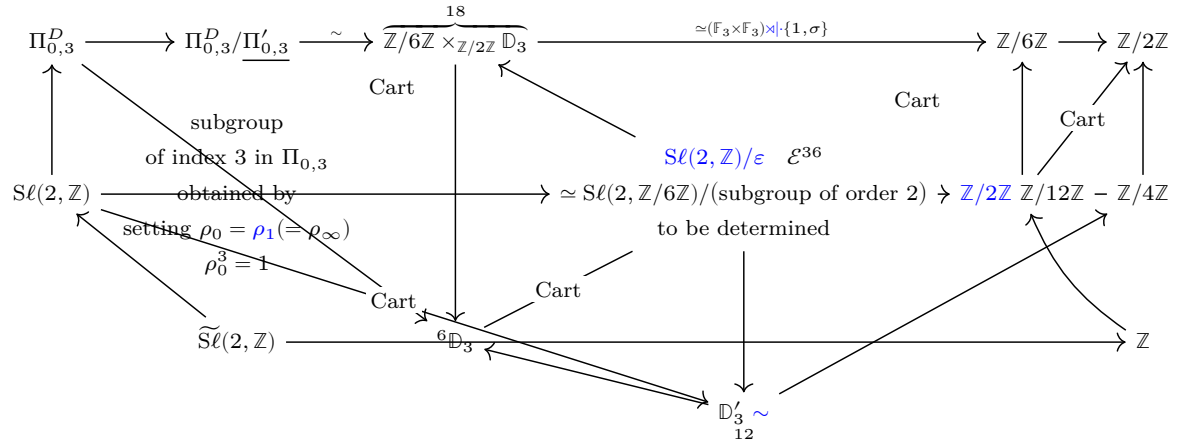
[scan 10]




$$\Pi^D_{0,3} = \{\rho, \sigma \mid \rho^3 = \sigma^2 = 1\}$$
$$S\ell(2,\mathbb{Z}) = \{\rho, \sigma \mid \underbrace{\rho^3 = \sigma^2}_{\omega},\ \rho^6 = 1\omega\ (= \sigma^4)\}$$

$$\Pi^D_{0,3}/\Pi'_{0,3} \simeq \{\rho, \sigma \mid \rho^3 = \sigma^2 = (\sigma\rho)^6 = 1,\ (\sigma\rho)^2 \text{central} = (\rho\sigma)^2\} \qquad 18$$
$$\downarrow \rho \mapsto 4 \bmod 6,\ \sigma \mapsto 3 \bmod 6 \qquad \overset{\rho \mapsto \rho}{\sigma \mapsto \sigma} \searrow$$
$$\mathbb{Z}/6\mathbb{Z} \qquad \mathbb{D}_3 = \{\rho, \sigma \mid \rho^3 = \sigma^2 2 = 1,\ \sigma(\rho) = \rho^{-1}\}$$
$$\text{i.e. } (\sigma\rho)^2 = 1$$

$$\mathcal{E} = \left\{ \rho, \sigma \;\middle|\; \underbrace{\rho^3 = \sigma^2}_{\omega},\ \rho^6 = 1\ (= \sigma^4),\ \rho \underbrace{(\sigma\rho)^2\omega}_{\rho_0 = \sigma\rho\sigma^{-1}\rho\ \omega} \text{ is invariant under } \rho \text{ and of order } \mid 3 \right\} \qquad 36$$

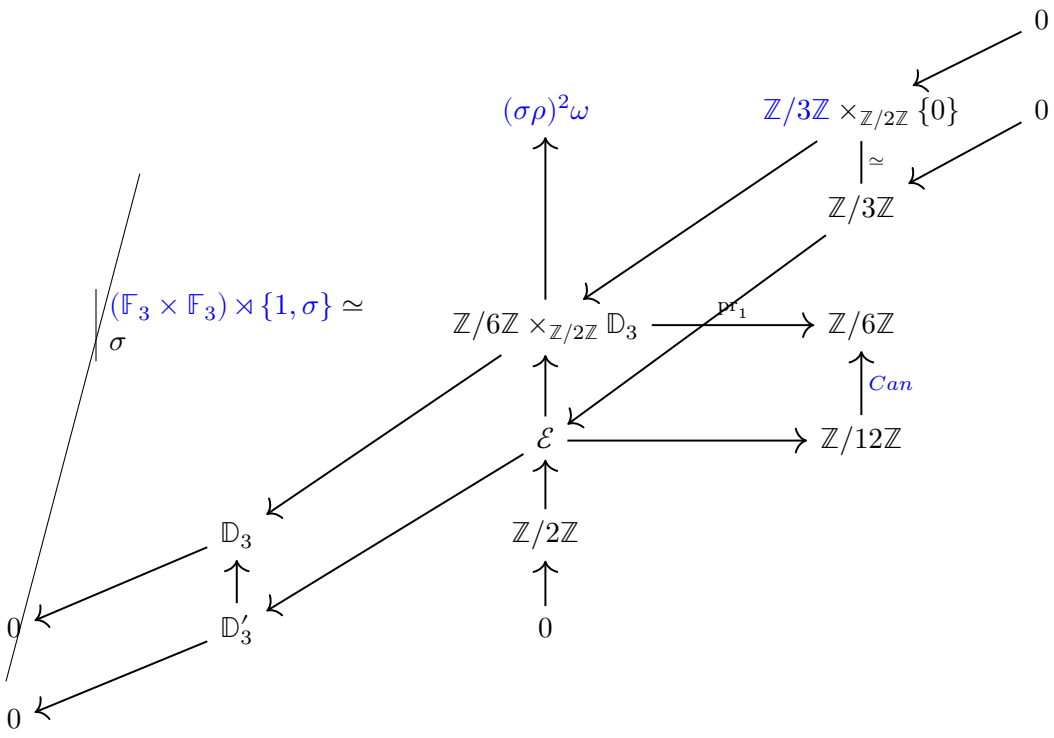
$$\text{i.e. } (\sigma\rho)^6 = \omega, \qquad (\sigma\rho)^2 = (\rho\sigma)^2$$

$$\begin{cases} \rho & \mapsto & 10|14 \bmod 12 \\ \sigma & \mapsto & 3 \bmod 12 \\ \omega & \mapsto & 6 \bmod 12 \end{cases}$$

$$\downarrow$$
$$\leftrightarrows\ \mathbb{Z}/12\mathbb{Z} \simeq (S\ell(2,\mathbb{Z}))_{\text{ab}}$$

$$\sigma\rho = \varepsilon_0 \longmapsto 1 \bmod 12$$
$$(\sigma\rho)^2\omega = \rho_0$$
$$(\sigma\rho)^2\omega = \rho_0 \longmapsto 8 \bmod 12$$
$$\mathbb{Z}/3\mathbb{Z}$$

$$\mathbb{D}'_3 = \left\{ \rho, \sigma \mid \rho, \sigma \mid \underbrace{\rho^3 = \sigma^2}_{\omega} = (\sigma\rho)^2,\ \rho^6 = 1\ (= \sigma^4) \right\} \qquad (\sigma\rho)^2\omega$$
$$\rho_0 \quad \longmapsto (2 \bmod 6, 1)$$
$$\| \quad (\sigma\rho)^2\omega.$$


0
$(\sigma\rho)^2\omega$
$\mathbb{Z}/3\mathbb{Z} \times_{\mathbb{Z}/2\mathbb{Z}} \{0\}$
0
$\simeq$
$\mathbb{Z}/3\mathbb{Z}$
$(\mathbb{F}_3 \times \mathbb{F}_3) \rtimes \{1, \sigma\} \simeq$
$\sigma$
$\mathbb{Z}/6\mathbb{Z} \times_{\mathbb{Z}/2\mathbb{Z}} \mathbb{D}_3$
$\mathrm{pr}_1$
$\mathbb{Z}/6\mathbb{Z}$
$Can$
$\mathcal{E}$
$\mathbb{Z}/12\mathbb{Z}$
$\mathbb{D}_3$
$\mathbb{Z}/2\mathbb{Z}$
0
$\mathbb{D}'_3$
0
0

9

[scan 12]

[p. 4]

---

*Note marginale.* ($n$ odd)

Proposition There exist liftings $\tilde{\rho}, \tilde{\sigma}$ of $\rho, \sigma$ such that we have $\tilde{\rho}^n = \tilde{\sigma}^2$ $\rho$, and the two sides are then of the form $\tilde{\omega}^\nu$, one of the for all, $\nu$ well-defined quantity|well-defined quantity modulo $2n$ (independently of the choices of the liftings $\tilde{\rho}, \tilde{\sigma}$) and $\nu \equiv \dfrac{n^2+1}{2}$ $\mathbb{Z}$, or :

$$\nu \bmod 2n \ \in \left\{ \left( -\frac{n-1}{2}, \frac{n+1}{2} \right) \right\}$$

and $\nu$ is odd. Which implies that there exists a $\nu_0$ unique in $\left\{ -\dfrac{n-1}{2}, \dfrac{n+1}{2} \right\}$ that is $\equiv \nu\ (2n)$. For $\nu$ fixed, $\tilde{\rho}, \tilde{\sigma}$ are uniquely determined. Thus we find|This provides presentation

$$\widetilde{\Pi}_n^D = \{ \tilde{\rho}, \tilde{\sigma}, \tilde{\omega} \mid \tilde{\omega} \text{ central}, \ \tilde{\rho}^n = \tilde{\sigma}^2 = \tilde{\omega}^{\nu_0 | \nu} \}.$$

$$\begin{array}{ccccc}
1 & & & & 0 \\
\downarrow & & & & \downarrow \\
\mathbb{Z}\tilde{\omega} & & \tilde{\omega} = (1, 2n) & & \mathbb{Z} \\
\downarrow & & & & \downarrow 2n \\
\widetilde{\Pi}_n^D & & \longrightarrow & & \mathbb{Z} \\
\downarrow & & & & \downarrow \\
\Pi_n^D = \{\rho, \sigma \mid \rho^n = \sigma^2 = 1\} & & \Longrightarrow & & (\Pi_n^D)_{\mathrm{ab}} \simeq \mathbb{Z}/2n\mathbb{Z} \\
\downarrow & & & & \\
1 & & & & \rho \longmapsto (n+1 \bmod 2n) = (1-n \bmod 2n) \\
 & & & & \sigma \longmapsto n \bmod 2n
\end{array}$$

$$\begin{array}{ll}
\tilde{\rho}_0 = (\rho, n+1) & \tilde{\rho}_0^n = (\overbrace{\widetilde{\rho^n}}^{1}, n(n+1)) = \tilde{\omega}^{n+1} \\
\tilde{\sigma}_0 = (\sigma, n) & \tilde{\sigma}_0^2 = (\underbrace{\sigma^2}_{1}, 2n)
\end{array}$$

$$\tilde{\rho}_0^n / \tilde{\sigma}_0^2 = (1, n(n-1)) = \tilde{\omega}^\alpha$$

$$\alpha = \frac{n-1}{2} \qquad (\alpha = 1 \text{ if } n = 3 ! | ?)$$

$$\left(\tilde{\rho}_0 \tilde{\omega}^\beta\right)^n = \left(\tilde{\sigma}_0 \tilde{\omega}^\gamma\right)^2 \qquad \text{i.e.} \quad \tilde{\rho}_0^n \tilde{\sigma}_0^{-2} = \tilde{\omega}^{n\beta - 2\gamma 2\gamma - n\beta}$$

$$\alpha + 1$$

$$n \qquad 2\gamma - n\beta = \alpha = \frac{n-1}{2}$$

$$4\gamma - 2n\beta =$$

$$\tilde{\rho} \overset{\text{definition}}{=} \tilde{\rho}_0 \tilde{\omega}^{\overset{\alpha}{\tilde{\beta}}} = \left( \rho, n+1+2n \underbrace{\overbrace{\frac{n-1}{2}}^{\beta=\alpha}}_{n(n-1)} \right) = (\rho, n^2+1)$$

$$\tilde{\sigma} \overset{\text{definition}}{=} \tilde{\sigma}_0 \tilde{\omega}^\gamma = \left( \sigma, 4n + 2n \underbrace{\underbrace{\underbrace{\gamma}_{\frac{(n-1)(n+1)}{4} = \frac{n^2-1}{4}}}_{\frac{n^3-n}{2}}}_{\frac{n^3+n}{2}} \right) = \left( \sigma, \frac{n^3-n}{2} \frac{n^3+n}{2} \right)$$

$$\left|\begin{array}{l}(\alpha+1).2-1.n=1\\ \underbrace{\alpha(\alpha+1)}_{\gamma}.2-\underbrace{\alpha}_{\beta}.n=\alpha\end{array}\right. \qquad \gamma =$$

*Note marginale.*

$$\left|\begin{array}{l}\tilde{\rho}=\tilde{\rho}_0\widetilde{\omega}^{2x}\\ \widetilde{\sigma}=\widetilde{\sigma}_0\widetilde{\omega}^{nx}\\ \\ x=\end{array}\right.$$

$$\tilde{\rho}_0^n=(1,n^3+n)=\tilde{\omega}^{\overbrace{\frac{n^2+1}{2}}^{\nu}}$$

$$\tilde{\sigma}_0^2=(1,n^3+n)=\tilde{\omega}^{\frac{n^2+1}{2}}$$

11

[scan 18]

[p. 6]

$S\mathcal{T}_{1,1} \simeq$ inverse image of $\mathrm{S}\ell(2,\mathbb{Z})$ in $\mathrm{G}\ell\mathrm{S}\ell(2,\mathbb{R})^{\sim}|\mathrm{G}\ell(2,\mathbb{R})^{\sim}$

$$\simeq \{\rho, \sigma \mid \rho^3 = \sigma^2 \ (= \omega)\}$$

---

*Note marginale.* (central elem$^{\text{ent}}$|el$^{\text{ement}}$

$$\mathcal{T}_{11} = S\mathcal{T}_{11}/\langle\omega^2\rangle$$

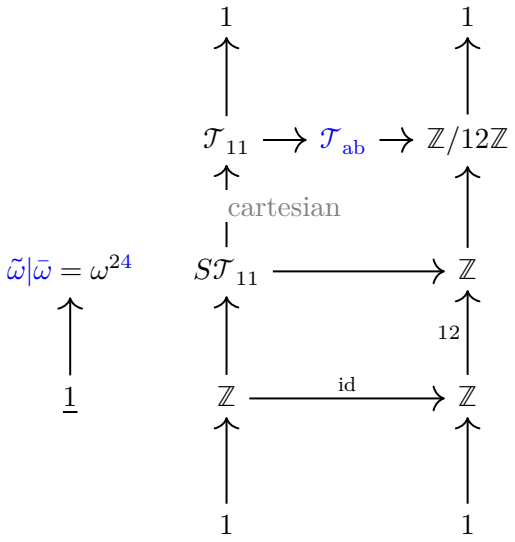
$(\mathcal{T}_{11})_{\text{ab}} \simeq \mathbb{Z}$ $\quad$ $(\mathcal{T}_{11})_{\text{ab}} \simeq \mathbb{Z}/12\mathbb{Z}$

$(\rho, \sigma)$

$\rho \longmapsto 2$ $\qquad$ $\rho \longmapsto 2 \bmod 12$

$\sigma \longmapsto 3$ $\qquad$ $\sigma \longmapsto 3 \bmod 12$

$\omega \longmapsto 6|8$

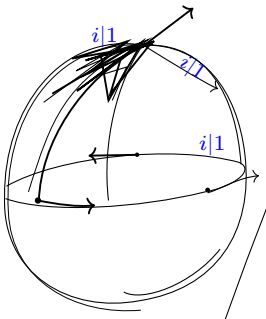



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*Note marginale.* so $S\mathcal{T}_{11}$ is obtained from $\mathcal{T}_{11}$ by inverse image of the canonical extension of $\mathbb{Z}/12\mathbb{Z}$ by $\mathbb{Z}$, via $\mathcal{T}_{11} \to \mathbb{Z}/12\mathbb{Z}$.

a normalization of $(K, K_0|N|k_0)$, then it is special : the category of étale coverings in $K'_fN|K'_{fN}$ – [illeg. 1] $\Delta$ covering|i.e. $\Delta$ topological has a unique normalization for which the projection is|be a morphism of normalized $\Delta$-complexes. [But beware, it works not for strict normalizations even|necessarily…]

8.|$\beta$. Fundamental|Fundamental groups|group and groupoids. We shall show|show the relation between topological construction and algebraic construction. 17




$\mathbb{Z}|\widetilde{\sigma}$ $\tilde{\rho}|\rho$ $\widetilde{\sigma}^2 = \tilde{\rho}^n = \bar{\omega}$

$\tilde{\rho}_0^n = \widetilde{\sigma}_0^n =$ $\tilde{\rho}_0\bar{\omega}|\bar{\omega}$

$\tilde{\rho}|\widetilde{\sigma}$ $\tilde{\rho} \longmapsto \tilde{\rho}\,\bar{\omega}^{-1} = \tilde{\rho}_0$

$\tilde{\rho}_0^n \longmapsto (\omega)|(\rho) = \tilde{\rho}^n\bar{\omega}^{-n} = \bar{\omega}^{1-n} = \omega^{\frac{1-n}{2}}$

$\widetilde{\sigma}_0 = \widetilde{\sigma}_{0|a}\bar{\omega}^s$ $\widetilde{\sigma}_0^{2|n} = \widetilde{\sigma}_{0|a}^2\bar{\omega}^{2s} = \bar{\omega}^{1+2s}$

$1 + 2s = \frac{1-n}{2}$ OK if $\frac{1-n}{2}$ odd

$R_i|R^i$ $\widetilde{R}_i = (e_j|e_g - e_{k|K})\lambda|h^{\omega}_{jk|gK}$

$t_\omega|\epsilon_\omega,\ h_\omega|h_1$

$V(\omega|w)$

$u\widetilde{\omega}^{,i}|\omega^i$

$P_i$ $H_i|H_2$

$\varphi_\omega(\widetilde{R}|\bar{R}_{K|k})$ $t_\omega|R_\omega$

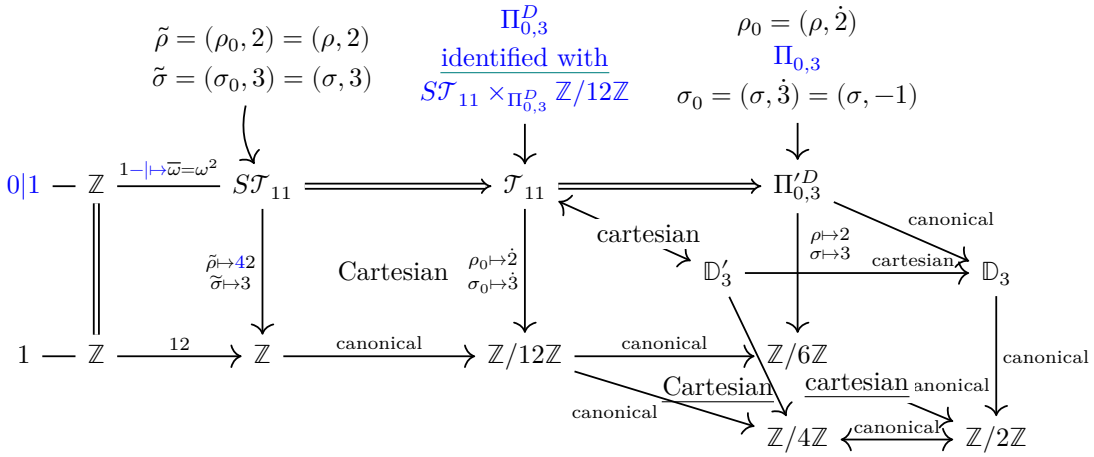
$\rho$ $\sigma$ act|acting on one of the paths|integers

distinguished

$R_K R|R_\kappa R$

[scan 20]

[p. 7]




*Note marginale.* for $\mathcal{T}_{11}$ identified with $\Pi^D_{0,3} \times_{\mathbb{Z}/2\mathbb{Z}} \mathbb{Z}/4\mathbb{Z}$

Have we indeed $\mid$ $S\mathcal{T}_{11} \simeq S\Pi^D_{0,3}$ ??
it would rather seem, however

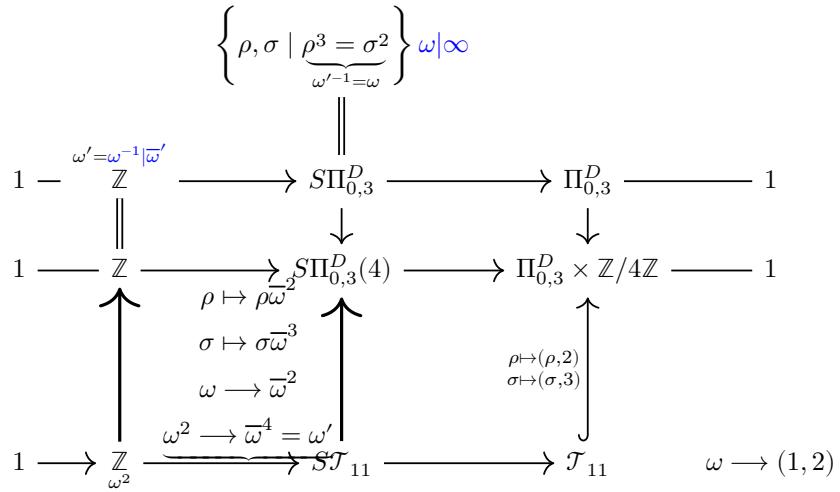
$$\mathcal{T}_{11} \quad PM_{11} \simeq \left( \underbrace{T^*U_{0,3}}_{\substack{\text{bundle} \\ \text{tangent}}}, \mathfrak{S}_3 \right)$$

(and not $PU_{0,3}$ !!)

$$\text{thus} \quad \mathcal{T}_{11} \simeq \pi_1(T^*U_{0,3}, \mathfrak{S}_3) \ \ldots$$

19

[scan 36]




---

*Note marginale.* $S\Pi^D_{0,3}(4) = \left\{\rho,\sigma,\overline{\omega} \mid \begin{matrix}\rho^3=\sigma^2=\overline{\omega}^{-4}\\ \overline{\omega}\text{ central}\end{matrix}\right\}$

N.B. $\dot{\overline{\omega}}|\overline{\omega}^4 = \omega^{-1}$
$= \omega'\omega$

$$\overbrace{S\Pi^D_{0,3}(4)}^{\pi_1(U_{0,3},\mathfrak{S}_3\times\mu_4)} \longrightarrow \Pi^D_{0,3}$$

35

[scan 302]

[p. 152]

tially to: determining the kernel of

$$SA|\mathcal{A}_{g,\nu}/(SA_{g,\nu})^0 \longrightarrow \Gamma^!_{g,\nu}, \quad \text{i.e.} \quad SA_{g,\nu} \cap A^0_{g,\nu}/(SA_{g,\nu})^0$$

as $T^I = T^{S_{g,\nu}}$. We can probably reduce to the context of compact surfaces with boundary (let us denote by a $'$ the corresponding topological groups), we have

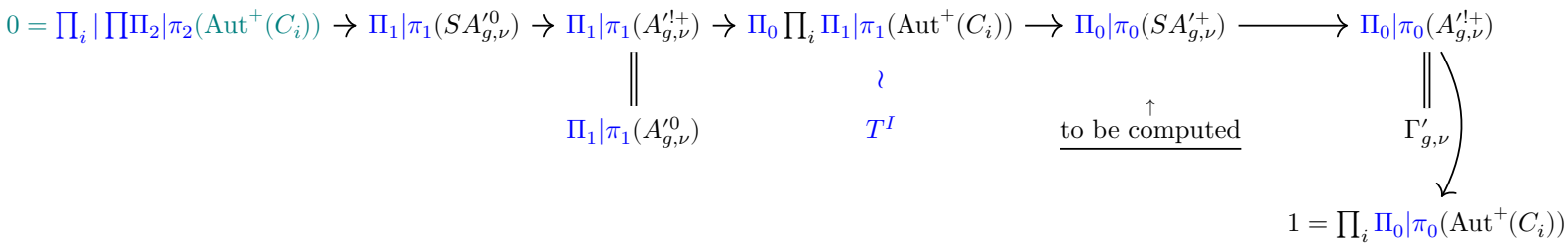
$$1 \longrightarrow SA'^+_{g,\nu} \longrightarrow A'^{!+}_{g,\nu} \longrightarrow \prod_{i \in S_{g,\nu}} \mathrm{Aut}^+(C_i) \longrightarrow 1$$

---

*Note marginale.* homotopic to the standard circle [illeg. 1] group $\mathbb{U}$, twisted by $T|\Pi$

where the $+$ indicates the automorphisms preserving the orientation, and the $C_i$ are the connected components of the boundary. We conclude from this the homotopy exact sequence

[Diagramme continué sur la page en paysage; 140-2-302-D1]


$0 = \prod_i | \prod \Pi_2 | \pi_2(\mathrm{Aut}^+(C_i))$
$\Pi_1 | \pi_1(SA'^0_{g,\nu})$
$\Pi_1 | \pi_1(A'^{!+}_{g,\nu})$
$\Pi_0 \prod_i \Pi_1 | \pi_1(\mathrm{Aut}^+(C_i))$
$\Pi_0 | \pi_0(SA'^+_{g,\nu})$
$\Pi_0 | \pi_0(A'^{!+}_{g,\nu})$
$\Pi_1 | \pi_1(A'^0_{g,\nu})$
$T^I$
to be computed
$\Gamma'_{g,\nu}$
$1 = \prod_i \Pi_0 | \pi_0(\mathrm{Aut}^+(C_i))$

$$\left[\text{and} \quad \Pi_i|\pi_i(SA'_{g,\nu}) \xrightarrow{\sim} \Pi_i|\pi_i(A'^0_{g,\nu}) \quad \text{if } i \geq 2\right]$$

i.e.

$$\begin{cases} 0 \longrightarrow \Pi_1|\pi_1(SA'^+_{g,\nu}) \longrightarrow \Pi_1|\pi_1(A'^0_{g,\nu}) \longrightarrow T^I \longrightarrow \Pi_0|\pi_0(SA'^{!+}_{g,\nu}) \longrightarrow \Gamma'^{!+}_{g,\nu} \longrightarrow 1. \\ \Pi_i|\pi_i(SA'^+_{g,\nu}) \xrightarrow{\sim} [\textit{illeg. 1}]\Pi_i|\pi_i(A'^0_{g,\nu}) \quad \text{for } i \geq 2 \end{cases}$$

In the case [illeg. 1]

We have an obvious homomorphism: (by 'gluing disks') $A'^0_{g,\nu} \longrightarrow A_{g,\nu}$ inducing $SA'_{g,\nu} \longrightarrow SA_{g,\nu}$, which and these are surely homotopy equivalences, so [illeg. 1] the preceding exact sequence

[p. 301]

[scan 328]

[p. 166|168]

$$(\overline{21})|(\overline{21}') \qquad 1 \longrightarrow \pi \longrightarrow \pi_1(U|\mathcal{U}_{g,\nu\overline{Q}}) \longrightarrow \pi_1(M_{g,\nu\overline{Q}}) \longrightarrow 1$$

$(\overline{22})$ $\quad 1 \longrightarrow \pi \longrightarrow$

We of course have an isomorphism of groups with loops|ends

$$(23) \qquad \pi \simeq \widehat{\Pi}_{g,\nu}$$

whose indeterminacy we would like to determine precisely

Let $M_{g,\nu C}$ denote the complex modular Teichmüller [illeg. 1]topos, defined using the differentiable$C^\infty$ surface $U|\mathcal{U}_{g,\nu}$, it is equipped with the canonical universal covering $\widetilde{M}_{g,\nu C}$ (§ 21)

---

*Note marginale.* $\widetilde{M}_{g,\nu C} = E_g/A^0_{g,\nu}|A^\circ_{g,\nu}$

with group of automorphisms $\mathcal{T}^+_{g,\nu}$ precisely –|... $U|\mathcal{U}_{g,\nu C}$ being itself also equipped with a universal covering over the preceding one, $\widetilde{U|\mathcal{U}}_{g,\nu C} = E_g/A^0_{g,\nu+1}|A^\circ_{g,\nu+1}$. This gives rise to the exact sequence of discrete groups (in the general topology context)

$$\begin{array}{lccccccc}
(24) & 1 \to \Pi_{g,\nu} & \longrightarrow & \mathfrak{S}|\mathcal{G}^+_{g,\nu} & \longrightarrow & \mathcal{T}^+_{g,\nu} & \longrightarrow & 1 \\
 & \downarrow & & \| & & \| & & \\
 & & & \pi_1(U|\mathcal{U}_{g,\nu C}) & & \pi_1(M_{g,\nu C}) & & \\
 & & & \downarrow & & \downarrow & & \\
(\text{mapping to} & 1 \to \Pi_{g,\nu} \to & \mathrm{Aut}_{\mathrm{lac}}\,\mathrm{ext}(\Pi_{g,\nu}) & \to & \mathrm{Aut}\,\mathrm{ext}_{\mathrm{lac}}(\Pi_{g,\nu}) & \to & 1 \\
 & & \| & & \| & & \\
 & & \mathfrak{S}|\mathcal{G}_{g,\nu} & & \mathcal{T}_{g,\nu} & &
\end{array}$$

by an isomorphism|homomorphism of extension structures), which by passage to the profinite completions gives the analogous exact sequence for the

[p. 327]

[scan 331]

is canonically endowed (not with a discrete oriented quasi-rigidification, which depends on the choice of an isomorphism $\overline{\mathbb{Q}} \simeq \overline{\mathbb{Q}}_0$ i.e. of an embedding $\overline{\mathbb{Q}} \subset \mathbb{C}$) but at least with a strict geometric Teichmüller group $\Sigma \subset \widehat{\mathcal{T}}(\pi)$, namely the image of $\pi_1(M_{g,\nu\overline{\mathbb{Q}}})$, isomorphic to This being so, We have a canonical homomorphism of group extensions

$$
\begin{array}{ccccccccc}
1 & \to & \pi_1(M_{g,\nu\overline{\mathbb{Q}}}) & \longrightarrow & \pi_1(M_{g,\nu}) & \longrightarrow & \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) & \longrightarrow & 1 \\
 & & \downarrow{\scriptstyle |\cdot|^2} & & \downarrow & & \downarrow & & \\
1 & \longrightarrow & \Sigma & \longrightarrow & \mathcal{N}|N_\Sigma & \longrightarrow & \Gamma_\Sigma & \longrightarrow & 1 \\
 & & & & \| & & & \searrow & \\
 & & & & \mathrm{Norm}_{\widehat{\mathcal{T}}(\pi)}(\Sigma) & & & & 1
\end{array}
\tag{26}
$$

*Note marginale.*

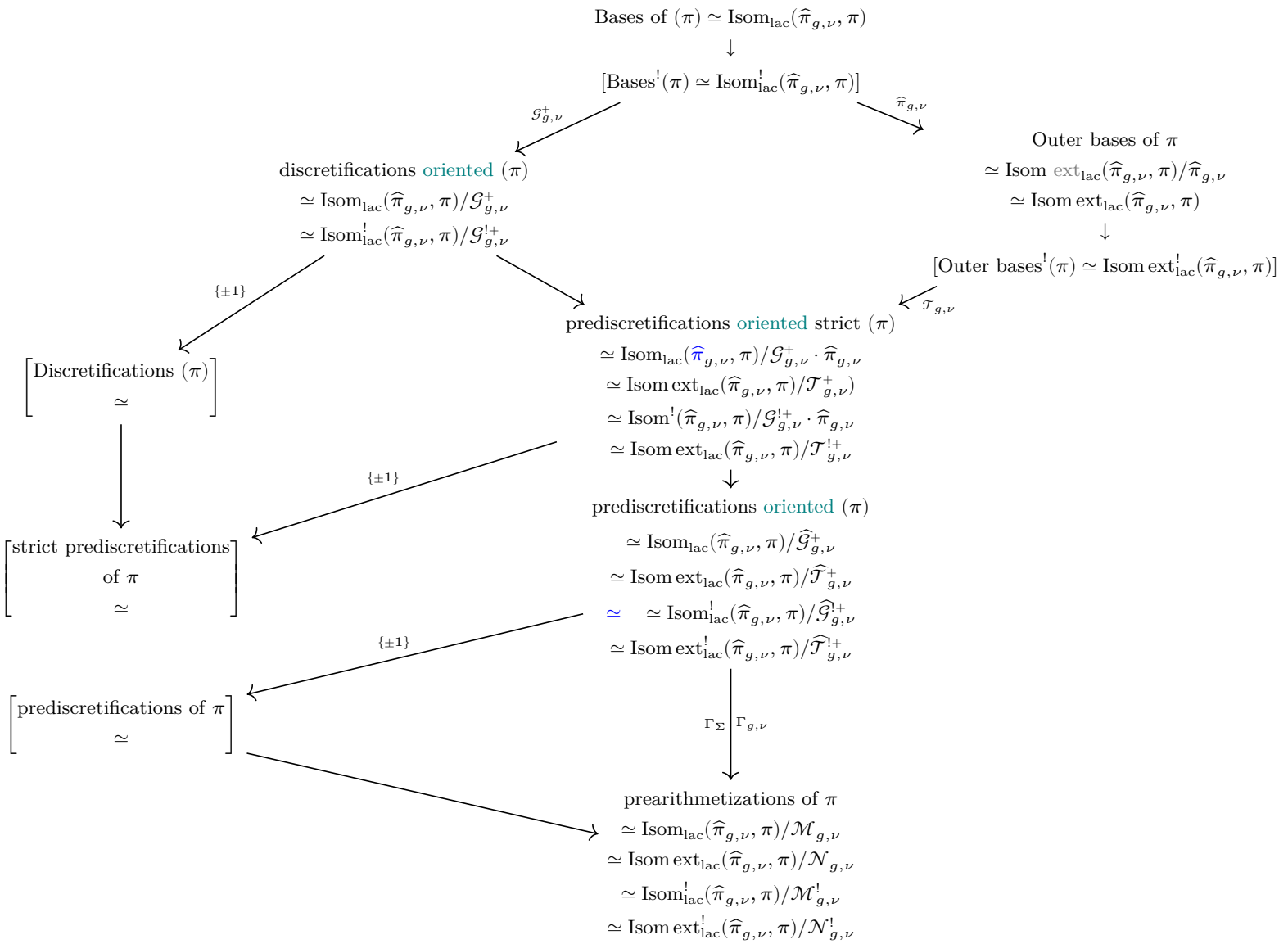
*Note marginale.* group of "arithmeticouter "arithmetic" automorphisms of $\pi$ (endowed with $\Sigma$)

For the choice of an isomorphism $\overline{\mathbb{Q}} \simeq \overline{\mathbb{Q}}_0$ i.e. of an embedding $\overline{\mathbb{Q}} \subset \mathbb{C}$, the image of complex conjugation in $\mathrm{Gal}(\overline{\mathbb{Q}}_0/\mathbb{Q})$ is none other than the element of order 2 of $\Gamma_\Sigma$ associated with the geometric Teichmüller subgroup, associated with the canonical quasi-discretification (oriented, but no matter) on $\pi$ (defined by $\overline{\mathbb{Q}} \simeq | \overset{\sim}{\to} \overline{\mathbb{Q}}_0$).

[scan 344]

[p. 174]

[Diagramme continué sur la page en paysage; 140-2-344-D1]


Bases of $(\pi) \simeq \mathrm{Isom}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)$
$[\mathrm{Bases}^!(\pi) \simeq \mathrm{Isom}^!_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)]$
$\mathcal{G}^+_{g,\nu}$
$\widehat{\pi}_{g,\nu}$
discretifications oriented $(\pi)$
$\simeq \mathrm{Isom}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{G}^+_{g,\nu}$
$\simeq \mathrm{Isom}^!_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{G}^{!+}_{g,\nu}$
Outer bases of $\pi$
$\simeq \mathrm{Isom}\ \mathrm{ext}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\widehat{\pi}_{g,\nu}$
$\simeq \mathrm{Isom}\,\mathrm{ext}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)$
$[\mathrm{Outer\ bases}^!(\pi) \simeq \mathrm{Isom}\,\mathrm{ext}^!_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)]$
$\{\pm 1\}$
$\mathcal{T}_{g,\nu}$
prediscretifications oriented strict $(\pi)$
$\simeq \mathrm{Isom}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{G}^+_{g,\nu} \cdot \widehat{\pi}_{g,\nu}$
$\simeq \mathrm{Isom}\,\mathrm{ext}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{T}^+_{g,\nu})$
$\simeq \mathrm{Isom}^!(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{G}^{!+}_{g,\nu} \cdot \widehat{\pi}_{g,\nu}$
$\simeq \mathrm{Isom}\,\mathrm{ext}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{T}^{!+}_{g,\nu}$
Discretifications $(\pi)$
$\simeq$
$\{\pm 1\}$
prediscretifications oriented $(\pi)$
$\simeq \mathrm{Isom}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\widehat{\mathcal{G}}^+_{g,\nu}$
$\simeq \mathrm{Isom}\,\mathrm{ext}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\widehat{\mathcal{T}}^+_{g,\nu}$
$\simeq \quad \simeq \mathrm{Isom}^!_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\widehat{\mathcal{G}}^{!+}_{g,\nu}$
$\simeq \mathrm{Isom}\,\mathrm{ext}^!_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\widehat{\mathcal{T}}^{!+}_{g,\nu}$
strict prediscretifications of $\pi$
$\simeq$
$\{\pm 1\}$
$\Gamma_\Sigma$
$\Gamma_{g,\nu}$
prediscretifications of $\pi$
$\simeq$
prearithmetizations of $\pi$
$\simeq \mathrm{Isom}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{M}_{g,\nu}$
$\simeq \mathrm{Isom}\,\mathrm{ext}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{N}_{g,\nu}$
$\simeq \mathrm{Isom}^!_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{M}^!_{g,\nu}$
$\simeq \mathrm{Isom}\,\mathrm{ext}^!_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{N}^!_{g,\nu}$

not separated

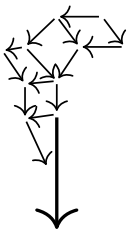
*Note marginale.*



*Note marginale.*




*Note marginale.*



[p. 343]

Please turn over

[scan 346]

[p. 175]

We observe that, with the exception of the two homogeneous spaces (under $\widehat{\widehat{\mathcal{G}}}(\pi)$, but not necessarily|necessarily under $\widehat{\widehat{\mathcal{G}}}(\pi, I')$, or even under $\widehat{\widehat{\mathcal{G}}}^!(\pi)$) Bases $(\pi)$ and Outer bases $(\pi)$, the other four are in fact homogeneous spaces under $\widehat{\widehat{\mathcal{G}}}^!$, and are expressed as quotients of the $\widehat{\widehat{\mathcal{G}}}^!_{g,\nu}$-torsor $\mathrm{Isom}^!(\hat{\pi}_{g,\nu}, \pi)$, by the four subgroups

$$\mathcal{G}^!_{g,\nu}, \quad \mathcal{G}^!_{g,\nu} \cdot \hat{\pi}_{g,\nu}, \quad \widehat{\mathcal{G}}^!_{g,\nu}, \quad \mathcal{M}^!_{g,\nu}$$

(the last defined as the inverse image of $\mathcal{N}^!_{g,\nu}$ in $\widehat{\widehat{\mathcal{G}}}_{g,\nu}$, just as $\mathcal{M}_{g,\nu}$ is defined as the inverse image of $\mathcal{N}_{g,\nu}$). We find on the other hand, if|for $I' = \{i_0, \dots, i_{\nu'-1}\}$, a|some homomorphism

$$\widehat{\widehat{\mathcal{G}}}^!_{g,\nu} \longrightarrow \widehat{\widehat{\mathcal{G}}}^!_{g,\nu'},$$

sending $\mathcal{G}^!_{g,\nu}$ into $\mathcal{G}^!_{g,\nu'}$, [illeg. 1] $\widehat{\mathcal{G}}_{g,\nu}$ into $\hat{\pi}_{g,\nu}$ by sending $\pi_{g,\nu}$ into $\pi_{g,\nu'}$, thus $\hat{\pi}_{g,\nu}$ into $\hat{\pi}_{g,\nu'}$, $\widehat{\mathcal{G}}^!_{g,\nu} = \Sigma^!_{g,\nu}$ into $\widehat{\mathcal{G}}^!_{g,\nu'} = \Sigma^!_{g,\nu'}$, and the subgroup $\mathcal{G}^!_{g,\nu} \cdot \hat{\pi}_{g,\nu}$ of [illeg. 1] $\widehat{\mathcal{G}}^!_{g,\nu}$ into the subgroup $\mathcal{G}^!_{g,\nu'} \cdot \hat{\pi}_{g,\nu'}$ of $\widehat{\mathcal{G}}^!_{g,\nu'}$. Finally, since $\Sigma^!_{g,\nu}$ maps into $\Sigma^!_{g,\nu'}$, [illeg. 1] $\mathcal{N}^!_{g,\nu}$ maps into the normalizer of $\Sigma^!_{g,\nu'}$ (in $\widehat{\widehat{\mathcal{T}}}^!_{g,\nu'}$), whichI say that this is none other than $\mathcal{N}^!_{g,\nu'}$. Changing notation, this amounts to the

Lemma The $\mathcal{N}^!_\Sigma = \mathrm{Norm}_{\widehat{\widehat{\mathcal{T}}}^!}(\Sigma^!)$ [illeg. 1] $(= \mathrm{Norm}_{\widehat{\widehat{\mathcal{T}}}}(\Sigma^!) \cap \widehat{\widehat{\mathcal{T}}}^!)$.

---

*Note marginale.* not proved [illeg. 4] of $\mathrm{Norm}_{\widehat{\widehat{\mathcal{T}}}}(\Sigma^!)/\Sigma^!$

Obviously we have $\mathcal{N}^!_\Sigma \subset \mathrm{Norm}_{\widehat{\widehat{\mathcal{T}}}}(\Sigma^!) \cap \widehat{\widehat{\mathcal{T}}}^!$, conversely let $g \in \widehat{\widehat{\mathcal{T}}}^!$ which normalizes $\Sigma^!$, let us show that it normalizes $\Sigma$ i.e. that it is $\in \mathcal{N}$ (thus $\in \mathcal{N}^! = \mathcal{N} \cap \widehat{\widehat{\mathcal{T}}}^!$). [p. 345]

[scan 348]

[p. 176]

i.e. $\widehat{\widehat{\mathcal{G}}}(\pi, i) \longrightarrow \widehat{\widehat{\mathcal{G}}}(\pi')$ is trivial on $\hat{\pi}$, and thus passes to the quotient as a $\widehat{\widehat{\mathcal{T}}}(\pi, i) \longrightarrow \widehat{\widehat{\mathcal{G}}}(\pi, i)$. I assume, that by analogy with the discrete case, that this homomorphism is an isomorphism, so that we have an exact sequence

---

*Note marginale.*



---

*Note marginale.* recall that we assume $\pi$ anabelian|nonabelian $(2g + \nu \geq 3)$

$$1 \longrightarrow \pi' \longrightarrow \widehat{\widehat{\mathcal{T}}}(\pi, i) \longrightarrow \widehat{\widehat{\mathcal{G}}}(\pi', i) \longrightarrow 1$$

or again

$$1 \longrightarrow \hat{\pi}_{g,\nu-1} \longrightarrow \widehat{\widehat{\mathcal{T}}}_{g,(\nu,\nu-1)} \longrightarrow \widehat{\widehat{\mathcal{G}}}_{g,\nu-1} \longrightarrow 1$$

which contains the exact sequence

$$1 \longrightarrow \pi^{*|+}_{g,\nu-1} \longrightarrow \mathcal{T}_{g,(\nu,\nu-1)} \longrightarrow \mathcal{G}_{g,\nu-1} \longrightarrow 1$$

as|known subparticular|dense part, sequence|is exact. It is convenient to work instead with

$$\begin{array}{ccccccccc}
1 & \text{——} & \pi_{g,\nu-1} & \text{——} & \mathcal{T}^{!+}_{g,\nu} & \longrightarrow & \mathcal{G}^{!+}_{g,\nu-1} & \text{——} & 1 \\
 & & \uparrow & & \uparrow & & \uparrow & & \\
1 & \text{——} & \hat{\pi}_{g,\nu-1} & \longrightarrow & \widehat{\mathcal{T}}^{!+}_{g,\nu} & \longrightarrow & \widehat{\mathcal{G}}^{!+}_{g,\nu-1} & \text{——} & 1 \\
 & & \| & & \uparrow & & \downarrow & & \\
1 & \text{——} & \hat{\pi}_{g,\nu-1} & \longrightarrow & \widehat{\widehat{\mathcal{T}}}^{!}_{g,\nu} & \longrightarrow & \widehat{\widehat{\mathcal{G}}}^{!}_{g,\nu-1} & \text{——} & 1
\end{array}$$

[illeg. 1]

The map [M1]

Prediscretifications|Prediscretifications$^{+}(\pi)$

$$\longrightarrow \text{Prediscretifications|Prediscretifications}^{+}(\pi')$$

is thus identified with $\widehat{\widehat{\mathcal{T}}}^{!}_{g,\nu}/\widehat{\mathcal{T}}^{!}_{g,\nu}$, we read off the diagram that this is $\simeq \widehat{\widehat{\mathcal{G}}}^{!+}_{g,\nu-1}/\widehat{\mathcal{G}}^{!+}_{g,\nu-1}$, whence bijectivity on the sets of oriented prediscretifications.

In the group $\widehat{\widehat{\mathcal{T}}}^{!}_{g,\nu} = \mathcal{T}$, we have an invariant subgroup $\hat{\pi}_{g,\nu-1} = \pi'$, and a subgroup $\Sigma$ $(= | \longleftarrow \widehat{\mathcal{T}}^{!+}_{g,\nu})$ [p. 347]

[scan 384]

property by saying that there exists a discretification $\pi_0$ of $\pi$, compatible with $a$, such that $Z \subset \mathcal{T}(\pi_0)$…)

Let us consider now an arbitrary finite subgroup $G \subset \widehat{\mathcal{T}}_a^+$. [N.B. If we take only $G \subset M_a$, so that the image of $G$ in $\mathbb{\Gamma}_a = M_a/\widehat{\mathcal{T}}_a^+$ is a finite subgroup, then if it $\mathbb{\Gamma}_a$ is true that $\mathbb{\Gamma}_a \simeq \mathcal{G}|\mathfrak{G}$, this image must be of order 1 or 2, and in the second case must define a $\tau \in \mathbb{\Gamma}_a$ corresponding to a well-determined prediscretification (not oriented) of $\pi$. This case should be reused again in what follows…]. If we Let us suppose, or admit, that $G$ is contained in $\mathcal{T}^+$,

$\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \longrightarrow$ ?

---

*Note marginale.* this is probably not automatic (if we do not|N' suppose $N_a = \mathrm{Norm}_{\widehat{\mathcal{T}}}(\widehat{\mathcal{T}}_a|\widehat{\mathcal{T}}_a^+)$)

for a suitable discretification in the class $a$, and that the outer action of $G$ thus obtained on a discrete $\pi_0$ of type $(g, \nu)$ is|are always realizable. It is therefore also realizable by|for an action of $G$

[scan 411]

[p. 208]

and unless I am mistaken, it has become obvious that this functor is an equivalence of categories (and as a consequence, there is the functor in the opposite direction : drill holes at "points" invariant under $\Gamma_{a|0}$ (or at arbitrary points, at the cost of passing to a smaller $\Gamma'$). for the isomorphisms

But I realize that [illeg. 1] this is not at all obvious – I am going to try to elucidate the situation axiomatically. We have fixed a genus $g$, we suppose given, for varying $\nu$ (such that $(g,\nu)$ anabelian) closed subgroups $N_{g,\nu} \subset \widehat{\widehat{\mathcal{T}}}_{g,\nu}$, normalizing $\widehat{\mathcal{T}}_{g,\nu}$, and satisfying the essential condition that the $\psi_i : \widehat{\widehat{\mathcal{T}}}_{g,\nu} \longrightarrow \widehat{\widehat{\mathcal{G}|\mathcal{S}}}_{g,\nu-1}$ induce

$$M_{g,\nu} \xrightarrow{\sim} M_{g,\nu-1}$$

Which makes it possible to define the notion of arithmetization of a profinite $\pi$ of anabelian type $(g,\nu)$, and the theory of

[scan 419]

[p. 212]

Thanks to this, the equivalence of categories between systems $(\pi, a, i, \Gamma, \theta : \Gamma \underset{\text{adm}}{\longrightarrow} N_a)$, and the systems $(\pi', a', \Gamma, \theta' : \Gamma \longrightarrow N_{a'}, P)$, where $P \in \mathrm{Pt}(B_{\pi', \Gamma^\natural})^\Gamma$, becomes perfectly precise at the level of points: the $\Gamma$-rational Points of $B_{\pi, \Gamma^\natural|, \Gamma^\natural}$ are identified with one|the set of the $\Gamma$-rational points of the $\Gamma$-rational points of $B_{\pi'}$, distinct from $P$.

This then allows us (which was not feasible in the discrete context!) to extend the equivalence of categories to an equivalence

(30)

| | |
|---|---|
| category of systems $(\pi, a\|\alpha, I', \Gamma, \theta : \Gamma \longrightarrow N_a)$ of an arithmetized $\pi$ (by $\alpha\|a$) of type $(g, \nu)$, of $I' \subset I(\pi)$, of an admissible outer action of $\Gamma$ on $\pi$ such that $\Gamma$ operates fixes each point of $I'$ | $\cong \mid \simeq$ category of systems $(\pi', a'\|\alpha', I', \Gamma^*\|\Gamma, \theta' : \Gamma \longrightarrow N_{a'})$ of an arithmetized $\pi'$ (by $a'\|\alpha'$) of type $(g, \nu')$ $[\nu' = \nu - \mathrm{card}(I')]$ with an admissible outer operation of $\Gamma$ on it, and a subset $I' \subset \mathrm{Pt}_{ad}(B_{\pi', \Gamma})$ (hence $I' \subset \mathrm{Pt}_{ad}(B_{\pi', \Gamma})^\Gamma$) i.e. $I'$ formed of points invariant\|fixed under $\Gamma$, i.e. "$\Gamma$-rational" |

*Note marginale.* with $2g + (\underbrace{\nu - \mathrm{card}\, I'}_{\nu'}) \geq 3$

We thus have a quasi-inverse functor: drilling holes at $I'$ !

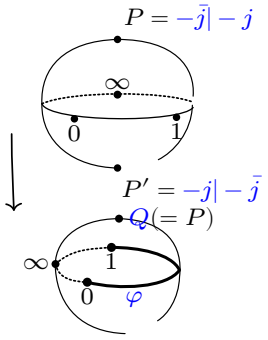
[scan 508]

[p. 258 (259)]

map determined on $X$ by the spherical triangles $(0, 1, \infty)$ (on the real axis as equator) – which is thus the universal weighted triangular map – as the inverse image of the universal map (having a single vertex 0, a single edge (folded) $0 \longrightarrow 1$, a single face, with center $\infty$).

---

*Note marginale.*




Every topological of étale covering of $U_{03}(\mathbb{C}) \subset X$ thus gives a topological étale covering of $U_{0,3}(\mathbb{C}) \subset Y$, ramified having [illeg. 1] over 1 ramification 2, over $\infty$ ramification 3 exactly|exactly, which corresponds to the "forgetful" functor (of the weighting), where a weighted triangular map is considered as just a map. From the point of view of fundamental groups we thus find an homomorphism of outer groups equivalence of categories

$$\begin{array}{ccc} \Pi_1(X(\mathbb{C}) \setminus \{0,1,\infty\}) & & \Pi_1\left(Y(\mathbb{C}) \setminus \{0,1,\infty\}; \underbrace{2\{1\} + 3\{\infty\}}_{\text{signature of ramification}}\right) \\ \| & \longleftarrow & \| \\ \Pi_{0,3} & & \Pi'_{0,3} \end{array}$$

$$\begin{array}{ccc} \text{([illeg. 1]4)} \quad \left(\text{Étale coverings of } X \setminus \{0,1,\infty\} = U_{0,3}\right) & \simeq & \left[\begin{array}{c}\text{Coverings of } Y \setminus \{0,1,\infty\} = U_{0,3} \\ \text{with ramification subordinate to} \\ \text{the signature } 2\{1\} + 3\{\infty\}\end{array}\right] \\ & & \text{/the object } (X \setminus \{0,1,\infty\}) \\ \wr & & \text{of the said category} \\ & & \wr \\ \Pi_{0,3}\text{-sets} & & (\Pi'_{0,3}\text{-Sets})/EE \end{array}$$

where $E$ is the $\Pi'_{0,3}$-set corresponding to the object $X \setminus \{0, 1, \infty\}$ of the category of étale coverings of $Y \setminus \{0, 1, \infty\}$ with ramification imposedsubordin-ate to $2\{1\} + 3\{\infty\}$

$\simeq \Pi'^{\circ}_{0,3} \in$ Ens

[scan 512]

above extend to this setting, and thus provide a homomorphism outer of extensions

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \widehat{\Pi}_{03} & \longrightarrow & \mathcal{E}_{03} & \longrightarrow & \Gamma & \longrightarrow & 1 \\ & & \downarrow & & \downarrow & & \| & & \\ 1 & \longrightarrow & \widehat{\Pi}'_{03} & \longrightarrow & \mathcal{E}'_{03} & \longrightarrow & \Gamma & \longrightarrow & 1 \end{array} \qquad (9)$$

, modulo conjugation by an inner automorphism of $\Pi'_{03}$.

---

*Note marginale.* (9 bis) $\widehat{\Pi}_{0,3} \longrightarrow \widehat{\Pi}'_{0,3}$ fits into an exact sequence

$$\boxed{1 \longrightarrow \widehat{\Pi}_{0,3} \longrightarrow \widehat{\Pi}'_{0,3} \longrightarrow \mathfrak{S}_3 \longrightarrow 1}$$

and ditto for the discrete groups

$$\boxed{1 \longrightarrow \Pi_{0,3} \longrightarrow \Pi'_{0,3} \longrightarrow \mathfrak{S}_3 \longrightarrow 1}.$$

That said, let $K$ be the kernel of the outer operation of $\Gamma$ on $\Pi'_{03}$. One then sees that $K$ operates the outer operation of $K$ on the open subgroup $\widehat{\Pi}_{03}$ takes place through a finite group (in fact, through the intermediary of $\widehat{\Pi}'_{03}/\widehat{\Pi}_{03} \simeq \mathfrak{S}_3$ ...) – which implies, by theorem 1 of the § above, that the group $K$ itself is finite. Thus by Artin one has $K = 1$ or $K = (1, \tau)$, $\tau$ complex conjugation. But since $\chi(\tau) = -1$ it is obvious that the outer operation of $\tau$ on $\Pi'_{03}$ (nor even on $(\Pi'_{03})^*_{\mathrm{ab}}$ [illeg. 2]) is not trivial... Which This proves the proposition.

$* \simeq \mathbb{Z}/2\mathbb{Z} + \mathbb{Z}/3\mathbb{Z}$

[scan 527]

[p. 269]

It is true that it is well known that a cohomology class (37) with torsion coefficients is killed, on passing to a finite extension of the $\mathbb{Q}$ base field ($\mathbb{Q}$ in this case). In fact, we have a spectral sequence

$$H^*(U_{0,3}, \mathbb{Z}_{/2}) \Longleftarrow E_2^{pq} = H^p(\overline{U}_{0,3}\mathbb{Q}, H^q(\overline{U}_{0,3}, \mathbb{Z}_{/2})|)) \tag{38}$$

and here $H^q(\overline{U}_{0,3}) = 0$ for $q \geq 2$, so we find an exact sequence

$$H^1(\mathbb{Q}) \longrightarrow H^1(U_{0,3}, \mathbb{Z}_{/2}) \longrightarrow H^1(\overline{U}_{0,3}, -)$$

$$\cdots \quad \underset{\|}{E_2^{0,1}} \longrightarrow \underset{\|}{E_2^{2,0}} \longrightarrow EH^2(U_{0,3}) \longrightarrow \underset{\|}{E_2^{1,1}} \longrightarrow \underset{\|}{E_2^{3,0}} \quad \cdots$$

$$H^0(\mathbb{Q}, H^1(\overline{U}) \qquad H^2(\mathbb{Q}, \mathbb{Z}_{/2}) \qquad H^1(\mathbb{Q}, H^1(\overline{U})) \underset{\mathbb{F}_2^2}{\wr} \qquad H^3(\mathbb{Q}, \mathbb{Z}_{/2}|\mathbb{F}_2)$$

I realize at last that the existence of a [illeg. 1] lifting, – i.e. of the hypothetical elliptic curve over $U_{0,3}$ – is completely trivial, it suffices to define it|it by the equation

$$y = \sqrt{x|\dot{x}(x-1)(x-\lambda)}$$

i.e.

$$y^2 - x(x-1)(x-\lambda) = F(x, y; \lambda] = 0$$

which defines an affine plane curve,

[scan 73]

[p. 325]

*Note marginale.*



$$(38) \qquad \underline{g\rho(g)\rho^2(g) = 1},$$

where $g$ is given by (40), and where $u$ is expressed in terms of $g$ $g$ hence of $g_0, \gamma$ by

$$(36) \qquad \begin{cases} u(\sigma_\infty) = \sigma_\infty \\ u(\rho) = g\rho \qquad \left( = \underbrace{\sigma_\infty(g_0)\ell_1^\gamma \rho(g_0)^{-1}}_{g}\,\rho \right) . \end{cases}$$

The relation (42)cf. $=$ above, then reads, since $L_0$ is generated by

$$\ell_0 = \underbrace{\ell_0'^{\,2}}_{\varepsilon_0} = (\sigma_\infty\rho)^2\sigma_\infty = \sigma_\infty\rho\sigma_\infty\rho$$

$$\underbrace{(u(\sigma_\infty\rho))}_{\sigma_\infty g\rho = g_0\sigma_\infty\ell_1^\gamma\rho g_0^{-1} = \mathrm{int}(g_0)(\sigma_\infty\ell_1^\gamma\rho)} \in \mathrm{int}(g_0)(L_0'|L_0) \qquad \text{now}$$

$$\mathrm{int}(g_0)(\sigma_\infty\ell_1^\gamma\rho)^2 g_0^{-1} = 1$$

thus (42) reads

$$(\sigma_\infty\ell_1^\gamma\rho)^2 \in L_0$$

i.e.

$$\underbrace{\underbrace{\sigma_\infty\ell_1^\gamma\sigma_\infty}_{\sigma_\infty(\ell_1)^\gamma}}_{\ell_0}\ \underbrace{\sigma_\infty[\mathit{illeg.}\ 1]\rho}_{\varepsilon_0}$$

N.B. $\mathrm{int}(g_0^{-1})u(\ell_0)$

*Note marginale.* N.B. we have $(\mathrm{int}(g_0^{-1})u)(\ell_0) = (\sigma_\infty\ell_1^\gamma\rho)^2 = \ell_0^{2\gamma+1}$

i.e.

$$\ell_0^\gamma\ \underbrace{\varepsilon_0^{2\gamma+1}}_{\varepsilon_0^{2\gamma}}\ \varepsilon_0 \in L_0$$

therefore …

$\mathrm{int}(g_0) \quad u(\ell_0) =$

The relation (42), or the apparently stronger relation

$$(43) \qquad u(L_0') \subset \mathrm{int}(g_0)(L_0')$$

reads

$$\underbrace{\underbrace{\mathrm{int}(g_0^{-1})}_{\sigma_\infty \ell_1^\gamma \rho = \underbrace{\sigma_\infty \ell_1^\gamma \sigma_\infty}_{\sigma_\infty (\ell_1)^\gamma} \underbrace{\sigma_\infty \rho}_{\sigma_\infty \rho \varepsilon_0}} \underbrace{u(\underbrace{\sigma_\infty}_{\ell_0' = \varepsilon_0} \rho)}_{\sigma_\infty g \rho = g_0 \sigma_\infty \ell_1^\gamma \rho g_0^{-1}}}_{\ell_0^\gamma = \varepsilon_0^{2\gamma}} \in L_0'$$

$$\varepsilon_0^{2\gamma+1}$$

---

*Note marginale.* N.B. we have $\mathrm{int}(g_0^{-1}) u(\underbrace{\ell_0'}_{\varepsilon_0}) = \varepsilon_0^{2\gamma+1}$

is satisfied automatically, we will have even

$$u(\ell_0') = u(\varepsilon_0) = \mathrm{int}(g_0)(\ell' \tag{44}$$

[scan 85]

it is generated by complex conjugation $\tau$

$$(5) \qquad , \quad \tau(z) = \bar{z},$$

the five other elements of $\mathcal{T}_{0,3}$ are

$$\tau\sigma \quad (6) \qquad \tau\dot{\sigma}_0 = \dot{\tau}_0, \quad \tau\dot{\sigma}_1 = \dot{\tau}_1, \quad \tau\dot{\sigma}_\infty = \dot{\tau}_\infty, \quad \tau\rho, \quad \tau\rho^{-1} = (\tau\rho)^{-1},$$

the anti-involutions|anti-involutions $\tau'_i$ $(i = 0, 1, \infty)$ are particularly interesting, given by

$$(7) \qquad \dot{\tau}'_0(z) = \frac{\bar{z}}{\bar{z} - 1}, \quad \dot{\tau}'_1(z) = \frac{1}{\bar{z}}, \quad \dot{\tau}'_\infty(z) = 1 - \bar{z}.$$

The fixed points in $U_{0,3}$ of the [illeg. 1] conformal automorphisms of [illeg. 2] $(\mathbb{P}^{1\mathrm{an}}, \{0, 1, \infty\})$ are the following

$$(8) \qquad \begin{cases} (\mathbb{P}^1_{\mathbb{C}})^{\dot{\sigma}_0} = \{0, 2\}, \\ (\mathbb{P}^1_{\mathbb{C}})^{\dot{\sigma}_1} = \{1, -1\}, \\ (\mathbb{P}^1_{\mathbb{C}})^{\dot{\sigma}_\infty} = \{\infty, \frac{1}{2}\} \\ \mathbb{P}^{1\,\dot{\rho}}_{\mathbb{C}} = \mathbb{P}^{1\,\dot{\rho}^{-1}} = \{-\bar{j}, -j\} \end{cases}$$

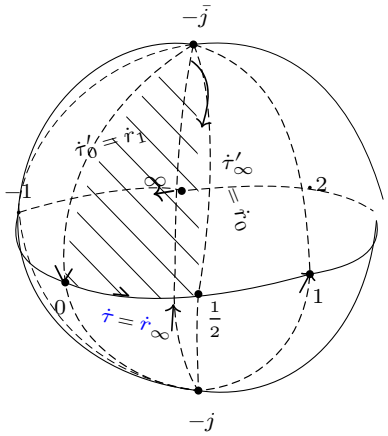
where $j = \exp 2i\pi/3 = \dfrac{-1 + i\sqrt{3}}{2}$ primitive root $3^{\text{ième|ème}}$ of 1

$-\bar{j} = \exp 2i\pi/6 = \dfrac{1 + i\sqrt{3}}{2}$ primitive root $6^{\text{ième|ème}}$ of 1

N.B. $\mathbb{P}^{1\,\dot{\rho}\tau}_{\mathbb{C}} = \mathbb{P}^{1\,\dot{\rho}^{-1}\tau}_{\mathbb{C}} = \emptyset$

$$(9) \qquad \begin{cases} \mathbb{P}^{1\,\tau}_{\mathbb{C}} = \mathbb{P}^1_{\mathbb{R}} \\ \mathbb{P}^{1\,\dot{\tau}'_0}_{\mathbb{C}} = \{x + iy \mid (x-1)^2 + y^2 = 1\} \\ \quad = \{z \mid z + \bar{z} = z\bar{z}\} = \frac{1}{z} \mid = \frac{1}{2} = \frac{1}{2} - \mathbb{U}, \\ \mathbb{P}^{1\,\dot{\tau}'_1}_{\mathbb{C}} = \{z \mid z\bar{z} = 1\} = \{x + iy \mid x^2 + y^2 = 1\} = \mathbb{U} \\ \mathbb{P}^{1\,\dot{\tau}'_\infty}_{\mathbb{C}} = \{z \mid z + \bar{z} = 1\} = \{x + iy \mid x = \frac{1}{2}\} = \frac{1}{2} + i\mathbb{R}; \end{cases}$$

The most suggestive picture is probably that of the Riemann sphere, where $\tau, \dot{\tau}'_0, \dot{\tau}'_1, \dot{\tau}'_\infty$ appear as the symmetriesreflections with respect to the corresponding "equators""gr$^{\text{eat}}$|great circles", which for $\tau$ is "the equator" and for the others are the great circles "meridians", passing through the poles $-\bar{j}, -j$, which effect the cutting up of the equator into six circular segments, by the subdivision points $0, \frac{1}{2}, 1, 2, \infty, -1$.



Let us also note the formulas

$$(10) \qquad \begin{cases} \dot\rho\dot\tau_0'\dot\rho^{-1} = \dot\tau_1', \quad \dot\rho\dot\tau_1'\dot\rho^{-1} = \dot\tau_\infty', \quad \dot\rho\dot\tau_\infty'\dot\rho^{-1} = \dot\tau_0' \\ \dot\tau_0'\dot\tau_1' = \dot\tau_1'\dot\tau_\infty' = \dot\tau_\infty'\dot\tau_0' = \dot\rho \end{cases}$$

84

[scan 88]

[p. 333]

We have

$$\begin{cases} (\sigma_\infty\rho)^2 = (\rho\sigma_1)^2 = \ell_0 = \varepsilon_0^2, & \varepsilon_0 \overset{\text{definition|definition}}{=} \sigma_\infty\rho = \rho\sigma_1 \overset{\text{dfn}}{=} \\ & \quad (\text{NB } \sigma_\infty = \rho\sigma_1\rho^{-1}) \\ (\sigma_0\rho)^2 = (\rho\sigma_\infty)^2 = \ell_1 = \varepsilon_1^2, & \varepsilon_1 = \sigma_0\rho = \rho\sigma_\infty \\ (\sigma_1\rho)^2 = (\rho\sigma_\infty|\sigma_0)^2 = \ell_\infty = \varepsilon_\infty^2, & \varepsilon_\infty = \sigma_1\rho = \rho\sigma_0 \end{cases} \tag{22}$$

The element $\varepsilon_i \in \mathfrak{S}_{0,3}$ (in fact $\varepsilon_i \in \mathfrak{S}^+_{0,3}$) is over $\dot\sigma_i$, it is characterized by the condition

$$\varepsilon_i^2 = \ell_i.$$

In the group $\mathcal{T}_{0,3}$ of order 12, there are seven involutive elements $\dot\sigma_0, \dot\sigma_1, \dot\sigma_\infty, \tau, \dot\tau_0', \dot\tau_1', \dot\tau_\infty'$, and four two elements of order 3 $\dot\rho, \dot\rho^{-1}$, and two of order 6 $\dot\rho\tau, \dot\rho^{-1}\tau$ (in addition to the identity).

Except for the last two (for which this is not possible*) We have lifted each of these elements to elements of the same order in $\mathfrak{S}_{0,3}$. We know that the possibilities, mod $\Pi_{0,3}$-conjugation, of|of the such liftings, correspond to the connected components of the fixed points in|of $U = U^{\text{an}}_{0,3}$ of the corresponding topological automorphisms. — for $\dot\sigma_0, \dot\sigma_1, \dot\sigma_\infty, \dot\tau_0', \dot\tau_1', \dot\tau_\infty'$ there is only one possibility mod conjugation (we have chosen $\sigma_0, \sigma_1, \sigma_\infty, \tau_0, \tau_1, \tau_\infty$ in the way that seemed most convenient for the calculations), for $\tau$ there are three — these are the three choices $\tau_0, \tau_1, \tau_\infty$ — for $\dot\rho, \dot\rho^{-1}$ there are two, corresponding|corresponding to the two poles $-\bar j, -j$ — we have chosen the one that corresponds to the pole $-\bar j$. The liftings corresponding to $-j$ can be deduced from them by conjugating by a $\tau_i$:

* indeed $\mathbb{P}^{1\,\dot\rho\tau} = \mathbb{P}^{1\,\dot\rho^{-1}\tau} = \emptyset$

87

[scan 107]

$$(81)\qquad \begin{cases} P=(a+b)(c+d-b) & Q=-a(c-b)-bc \overset{\text{if } \det g=+1}{=} 1\delta-a(c+d-b) \\ R=(c+d)^2-d(a+b) & S=ad-c(c+d)=\underset{\text{if } \det g=1}{\delta}-c(c+d-b) \\ \quad \underset{\text{if } \det g=+1}{=}(c+d)(c+d-b)-\delta & \end{cases}$$

where we have set $\delta = ad-bc = \det g$.

Comparing (81) and the expression (77) for $\rho^2(g^{-1})$, equation (79) – which is also written

$$g\rho(g) = \pm\rho^2(g^{-1})$$

gives the following solutions.

---

*Note marginale.* Up to replacing $g$ by $-g$, [illeg. 5] we are reduced to determining the solutions of the equation

$$(82)\qquad g|\rho\rho(g)\rho^2(g) = +|\pm 1 \qquad \text{i.e.} \quad g|\rho\rho(g) = +|\pm\rho^{-1}(g^{-1}) \qquad \underline{\text{in } \mathrm{Gl}(2,\widehat{\mathbb{Z}})}$$

(82) [illeg. 2] $g\rho(g) = +\rho^{-1}(g^{-1})$. We distinguish the solutions according to the factors of the product

$$\mathrm{Sl}(2,\widehat{\mathbb{Z}})' \simeq \prod_{\ell \text{ premier}} \mathrm{Sl}(2,\mathbb{Z}_\ell)'.$$

---

*Note marginale.* N.B. We shall use|use this study of equation (82) only in the case $\det g = 1$, i.e. $\delta = 1$, unless I am mistaken

The relation

$$\frac{1}{\delta}(a+b) = P = (a+b)(c+d-b) \qquad \text{in } \mathbb{Z}_\ell$$

gives the two solutions $a+b=0$ or $c+d-b=1$ (since $\mathbb{Z}_\ell$ is an integral domain).

1°) $(a+b) \neq 0$ (in $\mathbb{Z}_\ell$) whence $c+d-b=1$

$$(83)\qquad \boxed{c+d-b=\frac{1}{\delta}}.$$

Given (83), we verify at once that the formulas $g\rho(g) = \rho^{-1}(g^{-1})$ – which remain to be checked

$$P=(a+b)/\delta \quad Q=(1-a)/\delta$$
$$R=b/\delta \qquad S=(1-c)/\delta$$

[illeg. 1] $R =$

$$Q = \frac{c+d-a-b}{\delta}\left(=\frac{1}{\delta^2}|\frac{1}{\delta}-\frac{a}{\delta}\right),$$
$$R = \frac{b}{\delta}, \quad S = \frac{d-b}{\delta}\left(=\frac{1}{\delta^2}-\frac{c}{\delta}\right)$$

(84) are equivalent to [illeg. 5] (82) [illeg. 1]

$$(84)\qquad \boxed{\delta^3=1}.$$

---

*Note marginale.* The necessity of this formula (84) was obvious a priori.

2°) $a+b=0$ i.e. $a=-b$, i.e. $g=\begin{pmatrix}-b & b\\ c & d\end{pmatrix}$, and

$$-b(c+d)=1$$

$$(84)\quad \begin{cases} \det g=-b(c+d)=\delta \quad (\text{thus } \delta+b(c+d)=0;) \\ \qquad d=-\dfrac{\delta}{b}-c \\ P=0, \\ Q=\delta+b(c+d-b)=-b^2 \\ R=(c+d)^2=\dfrac{\delta^2}{b^2}, \\ S=ad \ [\textit{illeg. } 1] \\ \quad -c(c+d)+ad=1-c(c+d-b) \\ \quad [\textit{illeg. } 1]c=\delta+c\left(\dfrac{\delta}{b}+b\right) \end{cases}$$

[scan 320]

[p. 460]

of $\alpha$ as above, hence $\mu = \det \underline{u}$, hence and it suffices $\varepsilon\beta\sigma(\beta)^{-1} = \alpha|\eta$ remains to determine $\beta$ as $\beta = \underline{u}\underline{b}^{-1}$, with $\underline{b} \in \Gamma N'_\sigma$ arbitrary, $\varepsilon$ being defined then in terms of the choice of $\underline{b}$ as being $\varepsilon_\sigma(\underline{b})$. The condition for fixed $\alpha_0|\alpha$ to be completable by $(\beta, \mu, \varepsilon)$ to $(\alpha, \beta, \mu, \varepsilon) \in \Gamma\mathcal{G}^{\alpha|*}_{\rho,\sigma}$, is always the condition that the coefficient $D$ of $\zeta = \alpha+$

$$\zeta = \alpha\rho(\alpha)^{-1} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \tag{33}$$

be invertible, which makes it possible to will determine|determine $\mu$ by

$$\mu = 1/D. \tag{34}$$

Let us put

$$\alpha = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \tag{35}$$

and make $\alpha\rho(\alpha)^{-1}$ explicit; with $\rho = \begin{pmatrix} 0 & 1 \\ v & 1 \end{pmatrix}$ $\delta = \det\alpha = ad - bc$

$$\begin{aligned} \alpha\rho(\alpha)^{-1} &= \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 0 & 1 \\ v & w1 \end{pmatrix} \frac{1}{\delta} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \left(-\frac{1}{v}\right) \begin{pmatrix} w1 & -1 \\ -v & 0 \end{pmatrix} \\ &= -\frac{1}{\delta v} \begin{pmatrix} bv & a + bw \\ dv & c + dw \end{pmatrix} \begin{pmatrix} dw + bv & -d \\ -cw - av & c \end{pmatrix}, \end{aligned}$$

hence

$$= \frac{1}{\delta v}$$

$$\begin{cases} (-\delta v)A = vb(d + vb|bv) - (a + b)(c + va) & (-\delta v)B = -vbd + c(a + b) \\ (-\delta v)C = vd(d + vb) - (c + d)(c + va) & (-\delta v)D = -vd^2 + cd + c^2, \end{cases} \tag{36}$$

the most essential quantity thus being

$$\begin{cases} (-\delta v)D = -vd^2 + cd + c^2 = \det(d\rho + c), \\ \text{and after that|and after calculation where} \\ \quad \begin{cases} \alpha\ \delta = \det\alpha = ad - bc, \\ \quad v \text{ the parameter|coefficient which defines} \\ \quad \rho = \begin{pmatrix} 0 & 1 \\ v & 1 \end{pmatrix} \end{cases} \end{cases} \tag{37}$$

[scan 328]

so the condition reads $v|\nu = -1$, i.e. this is condition 3°, but then the expression (23) for $\rho^3$ shows us that $\rho^3 = -1$.

Remarks|Remark The preceding conditions imply the formulas

$$\begin{aligned} &(24) \qquad \varepsilon_0 = \sigma\rho \\ &(25) \qquad \varepsilon_1 = \rho\sigma \\ &(26) \qquad \rho\varepsilon_1 = \varepsilon_0^{-1} \\ &(27) \qquad \rho^2\sigma = \varepsilon_0^{-1} \end{aligned}$$

each of which except the last (27) implies them in turn — for example we have $\varepsilon_0 = -\sigma^{-1}\rho$ (7), so (24) is equivalent to $\sigma = -\sigma^{-1}$ i.e. $\sigma^2 = -1$, and (25) is equivalent to (24) by applying $\mathrm{int}(\rho)$ to both sides.

We have moreover

$$\rho\varepsilon_1 = \begin{pmatrix} 0 & 1 \\ v|\nu & w|\omega \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ v|\nu + w|\omega & w|\omega'|w|\omega \end{pmatrix}, \qquad \varepsilon_0^{-1} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

which shows that $\rho\varepsilon_1 = \varepsilon_0^{-1}$ is equivalent to:

$$w|\omega = 1, \quad v|\nu + w|\omega = 0 \qquad \text{i.e.} \quad w|\omega = 1, v|\nu = -1 \quad \text{i.e.} \quad \rho = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix},$$

OK. Composing (25) (in the form $\rho\varepsilon_1 = \rho^2\sigma$) and (26) we find (27) as a consequence of the conditions 2°) + 3°); theLet us try to prove converse implication followsby from a brute-force calculation

$$\rho^2\sigma = \rho^2\sigma = \begin{pmatrix} -v|\nu w|\omega & w|\omega - (v|\nu + w|\omega^2) \\ -v|\nu(v|\nu + w|\omega^2) & -w|\omega(w|\omega^2 + 2v|\nu) + v|\nu - w|\omega^2 \\ & |\, - w|\omega(w|\omega^2 + 2v|\nu) + v|\nu + w|\omega^2 \end{pmatrix},$$

$$\varepsilon_0^{-1} = \begin{pmatrix} 1 & -1/v|\nu \\ 0 & 1 \end{pmatrix}$$

since $v|\nu$ invertible we can have the equality $\rho^2\sigma = -\varepsilon_0^{-1}|\varepsilon_0^{-1}$ only if $-v|\nu(v|\nu + w|\omega^2) = 0$, this implies|which implies ($v|\nu$ invertible!) $v|\nu + w|\omega^2 = 0$, whencei.e. $v|\nu = -w|\omega^2$,

[p. 327]

[scan 329]

[p. 465]

$$\rho^2\sigma = \begin{pmatrix} w|\omega^3 & w|\omega \\ 0 & -w(1+2w)w|\omega^3 \end{pmatrix},$$

This shows that $\rho^2\sigma = \varepsilon_0^{-1}$ iff $w|\omega = 1$ OK.

---

*Note marginale.* We have in addition to [illeg. 4] $v|\nu = -w|\omega^2$ [illeg. 2] i.e. $v|\nu = -1, w|\omega = 1$

$$(27\text{bis}) \qquad \rho^2\sigma = \varepsilon_0^{-1} \iff \{v|\nu + w|\omega^2 = 0,\ w|\omega^3 = 1$$
$$(\Rightarrow v|\nu = -w|\omega^2 = -1/w|\omega \text{ i.e. } w|\omega = -1/v|\nu,\ \text{OK})$$

But (27), on multiplying by $\rho$, is also equivalent to

$$(27)\ [illeg.\ 1] \rho^3\sigma = u\rho\varepsilon_0^{-1}$$

now we know that $\varepsilon_0 = -\sigma^{-1}\rho$ (7) i.e. $\varepsilon_0^{-1} = -\sigma\rho^{-1}\sigma|\sigma_0$ i.e. $\rho\varepsilon_0^{-1} = -\sigma$, so the preceding relation also reads $\rho^3\sigma = -\sigma$ or again

$$(28) \qquad \boxed{\rho^3 = -1}.$$

We can see this the equivalence directlyof (28) with the conditions (27bis) by computing $\rho^3$ in terms of $v|\nu, w|\omega$, we find

$$(29) \qquad \rho^3 = \begin{pmatrix} v|\nu w|\omega & v|\nu + w|\omega^2 \\ v|\nu(v|\nu + w|\omega^2) & w|\omega(2v|\nu + w|\omega^2) \end{pmatrix}$$

And we can have $\tilde{\rho}^3 = 1$ i.e. $\rho^3$ a scalar only if $v|\nu + w|\omega^2 =$

$$v|\nu + w|\omega^2 = 0 \qquad \text{i.e.} \quad v|\nu = -w|\omega^2$$

which implies

$$\rho^3 = \begin{pmatrix} -w|\omega^3 & 0 \\ 0 & -w|\omega^3 \end{pmatrix}$$

i.e. we find

Lemma We have $\tilde{\rho}^3 = 1$ iff $v|\nu + w|\omega^2 = 0$ i.e. $v|\nu = -w|\omega^2$ i.e. we have

$$(30) \qquad \rho = \begin{pmatrix} 0 & 1 \\ -w|\omega^2 & w|\omega \end{pmatrix},$$

and then we have

$$(31) \qquad \rho^3 = -w|\omega^3 \cdot 1\ ,$$

so we have

$$(32) \qquad \rho^3 = -1 \iff v|\nu = -w|\omega^2 \qquad \begin{matrix} w|\omega^3 = 1 \text{ i.e. } w|\omega \in \Gamma\mu_3 \\ \underline{and} \quad v|\nu = -w|\omega^2\ (= -1/w|\omega) \end{matrix}$$

[p. 328]

XI Recapitulation on the group $\mathcal{M}_{\rho,\sigma}$. 

Ⓛ Given, over a scheme $S$, a form $G$ of $\mathrm{Gl}(2)$, hence forms $SG$, $PG$ of $\mathrm{Sl}(2)$, GP(1)|PGl(1), and two sections $\tilde{\rho}, \tilde{\sigma}$ of $PG$, and a 1-parameter unipotent|multiplicative additive bordered|based subgroup

$$\underset{\sim}{\tilde{\varepsilon}}_0 : G_a \hookrightarrow PG$$

$\varepsilon_0|G_a$ (which amounts to the same thing, a of regular unipotent section of $\varepsilon_0$ of $G$, or a regular nilpotent section $N_0$ in the form $\mathcal{M}$ of $\mathrm{M}_2(\mathcal{O}_S)$ associated to $G$) — satisfying the conditions

$$(1)\quad \tilde{\sigma}^{-1}\tilde{\rho} = \underset{\sim}{\tilde{\varepsilon}}_0(1), \qquad \underset{\sim}{\tilde{\varepsilon}}_0 \text{ and } \underset{\sim}{\tilde{\varepsilon}}_1 = \rho(\underset{\sim}{\tilde{\varepsilon}}_0)(= \sigma(\underset{\sim}{\tilde{\varepsilon}}_0)) \text{ are independent on each fiber}$$

$$(2)\quad \mathrm{Tr}\,\sigma = 0 \qquad \text{if } \sigma \text{ is a local lifting of } \tilde{\sigma},$$

We define canonical liftings $\rho$, $\sigma$ of $\tilde{\rho}, \tilde{\sigma}$ to $G$, satisfy by the conditions

$$(3)\quad \sigma\rho \begin{cases} \varepsilon_0 = -\sigma^{-1}\rho & \text{i.e. } \varepsilon_1 = -\rho\sigma^{-1} \\ \mathrm{Tr}(\rho+\sigma) = 1 & \text{i.e. } \mathrm{Tr}\,\rho = 1, \end{cases}$$

and the constructions which follow make sense.

We construct a pinning

$$(4)\ (5)\quad \tau_{\rho,\sigma} = \mathrm{Gl}(2) \xrightarrow{\sim} G$$

by the condition that $\varepsilon_1$ take the form

$$(6)\qquad \varepsilon_1 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

whence

$$(7)\quad \begin{cases} \rho = \begin{pmatrix} 0 & 1 \\ v|v & 1 \end{pmatrix}, \qquad \sigma = \begin{pmatrix} 0 & -1 \\ -v|v & 0 \end{pmatrix} \qquad (v|v \in \Gamma G_m \\ \det\rho = \det\sigma = -v|v \end{cases}$$

$$(8)\qquad \varepsilon_0 = \begin{pmatrix} 1 & 1/v|v \\ 0 & 1 \end{pmatrix},$$

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[scan 425]

[p. 514]

$$\mathcal{R}^{\circ}_{\rho,\sigma} \simeq \left\{ \left( j_{T_0}(\mu^2),\ \underset{\Gamma T_\rho}{\underset{\cup}{\underline{a}}}\ ,\ \underset{\Gamma T_\sigma}{\underset{\cup}{\underline{b}}} \right) \cdot \underbrace{\mu j_G(\mu^{-2})}_{\substack{\text{N.B. the determinant is 1} \\ \text{since } \det j_G(\mu^{-2}) = \mu^{-2}}} \;\middle|\; \begin{array}{c} \mu \in \Gamma\mathbb{G}_m, \\ \det \underline{a} = \det \underline{b} = \lambda^2 \\ \underline{a} \in \Gamma T_\rho, \qquad = \mu^2 \\ \underline{b} \in \Gamma T_\sigma, \end{array} \right\} \tag{71}$$

where $j^+|j_G$

$$\begin{cases} j(\mu) = \begin{pmatrix} \mu & 1 \\ 0 & 1 \end{pmatrix} \in \Gamma SG \qquad (\mu \in \Gamma\mathbb{G}_m) \\ j_{T_0}(\mu) : \mathbb{G}_m \longrightarrow T_0 \quad \text{given by } j \\ j_G : \mathbb{G}_m \longrightarrow SG \quad \text{given by } j \text{ above.} \end{cases} \tag{72}$$

We thus find that the composite

$$\begin{cases} \mathcal{R}^{\circ}_{\rho,\sigma} \underset{\text{inclusion}}{\hookrightarrow} \mathcal{M}^{\circ}_{\rho,\sigma} \underset{\text{canonical}}{\longrightarrow} \begin{array}{c} \mathcal{T}^{\circ}_{\rho,\sigma} \simeq T_\rho \times_{\mathbb{G}_m} T_\sigma \\ \| \\ \mathcal{M}^{\circ}_{\rho,\sigma}/\widetilde{\mathcal{G}}^{+}_{\rho,\sigma} \end{array} \\ \text{makes } \mathcal{R}^{\circ}_{\rho,\sigma} \text{ an algebraic covering} \\ \text{of } \mathcal{T}^{\circ}_{\rho,\sigma} \end{cases} \tag{73}$$

and more precisely that we have a cartesian diagram

$$\begin{array}{ccc} \mathcal{R}^{\circ}_{\rho,\sigma} & \xrightarrow{(73)} & T_\rho \times_{\mathbb{G}_m} T_\sigma \simeq \mathcal{T}^{\circ}_{\rho,\sigma} \\ \Big\downarrow{\scriptstyle \text{the canonical homomorphism of (70)}} & \text{cartesian} & \Big\downarrow{\scriptstyle \text{canonical}} \\ \mathbb{G}_m & \xrightarrow[\mu \longmapsto \mu^2]{} & \mathbb{G}_m \end{array} \tag{74}$$

(75) Proposition : $\mathcal{R}^{\circ}_{\rho,\sigma} = \underline{Centr}(\mathcal{M}^{\circ}_{\rho,\sigma})$

Obviously the center of $\mathcal{M}^{\circ}_{\rho,\sigma}$ is contained in $\mathcal{R}^{\circ}_{\rho,\sigma}$, precisely the inverse image of the center

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[scan 446]

$\underline{\mathcal{O}}_S$. Let us denote by $\mu_2'$

$$\begin{array}{c} \mu_2' \subset ST_\sigma \\ \| \\ \underline{\mathrm{Cent}}(ST_\sigma)^{\mu_2} \end{array} \tag{131}$$

this|a subgroup of

whence finally|finally, namely|i.e.:

$$\underline{\mathrm{Cent}}(\mathcal{M}_{\rho,\sigma}) \simeq S'T_\rho \times \mu_2 \times G_m \subset R_{\rho,\sigma} \simeq S'T_\rho \times SN'_\sigma \times G_m \tag{128}$$

$=$

whence

$$\underline{\mathrm{Cent}}(\mathcal{M}_{\rho,\sigma}) \cap \mathcal{M}^\circ_{\rho,\sigma} = ST_\rho \times \mu_2 \times G_m \subset R^\circ_{\rho,\sigma} \simeq ST_\rho \times ST_\sigma \times G_m \tag{129}$$

Let us return to the integral sections in the absolute modular case

$$R_{\rho,\sigma}(\mathbb{Z}) \simeq \underbrace{S'T_\rho(\mathbb{Z})}_{T_\rho(\mathbb{Z}) \simeq \mathbb{Z}/6\mathbb{Z}} \times \underbrace{SN'_\sigma(\mathbb{Z})}_{T_\sigma(\mathbb{Z}) = \mathbb{Z}/4\mathbb{Z}} \times \underbrace{G_m(\mathbb{Z})}_{\mu_2(\mathbb{Z}) \simeq \pm 1} \tag{130}$$

$$\begin{aligned} \underline{\mathrm{Cent}}(\mathcal{M}_{\rho,\sigma})[\mathbb{Z}]|(\mathbb{Z}) &\simeq S'(T_\rho[\mathbb{Z}]|(\mathbb{Z})) \times \mu_2(\mathbb{Z}) \times G_m(\mathbb{Z}) \\ &\simeq \mathbb{Z}/6\mathbb{Z} \times \{\pm 1\} \times \{\pm 1\} \end{aligned} \tag{131}$$

and for the record (cf. (75), p. 514)

$$\begin{aligned} \underbrace{\underline{\mathrm{Cent}}(\mathcal{M}^\circ_{\rho,\sigma})}_{\substack{R^\circ_{\rho,\sigma} \\ \simeq ST_\rho \times ST_\sigma \times G_m}}(\mathbb{Z}) &\simeq ST_\rho(\mathbb{Z}) \times ST_\sigma(\mathbb{Z}) \times G_m(\mathbb{Z}) \\ &\simeq \mathbb{Z}/6\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z} \times \{\pm 1\}. \end{aligned} \tag{132}$$

In summary, we have the diagram of strict inclusions of group schemes

$$\begin{array}{lcl} \text{dimension 3} & \underline{\mathrm{Cent}}(\mathcal{M}^\circ_{\rho,\sigma}) = R^\circ_{\rho,\sigma} \hookrightarrow R_{\rho,\sigma} & \text{dimension 3} \\ & \simeq ST_\rho \times ST_\sigma \times G_m \overset{(\text{quotient } \mu_2 \times \mu_2)}{\simeq} S'T_\rho \times SN'_\sigma \times G_m & \\ & \uparrow \qquad\qquad\qquad \uparrow & \\ \text{dimension 2} & \underline{\mathrm{Cent}}(\mathcal{M}_{\rho,\sigma}) \cap \mathcal{M}^\circ_{\rho,\sigma} \hookrightarrow \underline{\mathrm{Cent}}(\mathcal{M}_{\rho,\sigma}) & \text{dimension 2} \\ & = ST_\rho \times \mu_2 \times G_m \overset{(\text{quotient } \mu_2)}{\simeq} S'T_\rho \times SN'_\sigma\mu_2 \times G_m) & \end{array} \tag{133}$$

[p. 445]



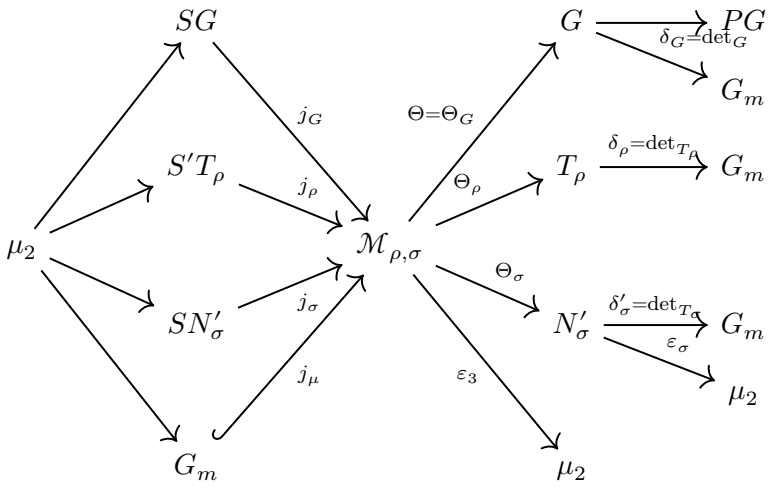
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[scan 448]

*Note marginale.* Where we define $\mathcal{M}_{\rho,\sigma}$

℗ Let us make a list|little recapitulation|recapitulatory of the main group homomorphisms considered in the course of section 6°, involving the group $\mathcal{M}_{\rho,\sigma}$

(134)



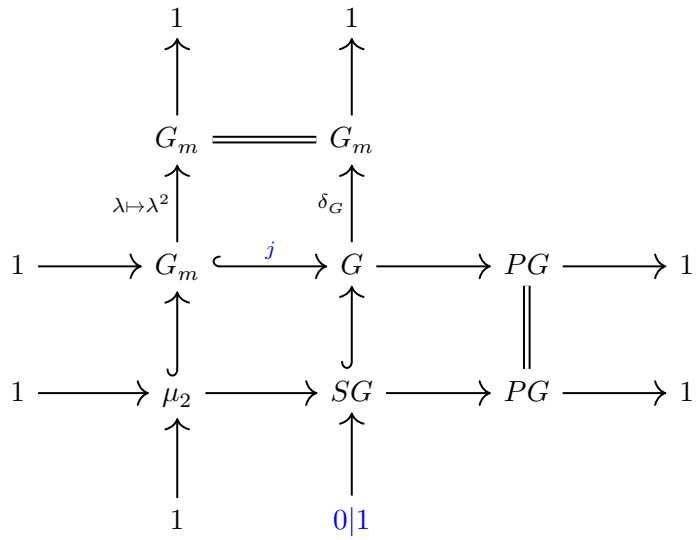

Let us note that this diagram can be reconstructed (with the essential properties which we are going to pass in review) from the data

(135)

$$\begin{cases} G_m \xrightarrow{i_\rho} T_\rho \xrightarrow{\delta_\rho} G_m & (\delta_\rho i_\rho(\lambda) = \lambda^2) \\ G_m \xrightarrow{i_\sigma} N'_\sigma \xrightarrow{\delta_\sigma} G_m & (\delta_\sigma i_\sigma(\lambda) = \lambda^2) \\ N'_\sigma \xrightarrow{\varepsilon_\sigma} \mu_2 & \varepsilon_\sigma i_\sigma(\lambda) = 1 \\ \begin{array}{ccccccccc} 1 & \longrightarrow & G_m & \longrightarrow & G & \longrightarrow & PG & \longrightarrow & 1 \\ & & \uparrow & \text{cartesian} & \uparrow & & \| & & \\ 1 & \longrightarrow & \mu_2 & \longrightarrow & SG & \longrightarrow & PG & \longrightarrow & 1 \end{array} & \end{cases}$$

in addition to the familiar datum

(136)



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[scan 548]

[p. 577]

It is ultimately this element, which normalizes both $L_\rho$ and $L_\sigma$ by via the remarkable formulas

$$(*) \quad \tau'(\rho) = \rho^{-1}, \qquad \tau'(\sigma) = \sigma^{-1}$$

that takes precedence over $\tau$, which satisfies

$$(**) \quad \tau(\varepsilon_0) = \varepsilon_0^{-1}, \qquad \tau(\varepsilon_1) = \varepsilon_1^{-1} \quad \text{and even } \tau(\sigma) = \sigma^{-1}$$

(which is obviously very virtuous !), but whose action on $\rho$

$$\tau(\rho) = \sigma(\rho^{-1})$$

is less convenient.

---

*Note marginale.* Nevertheless $\tau'$ has over $\tau$ the serious disadvantage of not being an element of the group $\widehat{\Pi}'_{0,3} = \{1, \tau\}.\widehat{\Pi}_{0,3}$, where we have the rigidity relation $\widehat{\Pi}'^{\sigma}_{0,3} = \widehat{\Pi}'^{\rho}_{0,3} = \{1\}$ — whereas in $\{1, \tau'\}.\widehat{\Pi}_{0,3}$, the centralizer of $\sigma$ is not reduced to 1, but is equal to $\{1, \tau'\}$ Moreover, $\tau'$ has the disadvantage of not being in $\mathcal{M}^{!\prime}_{0,3}$ ...

## V

Preliminary data

1°) So let $\mathcal{G}$ be a profinite group, equipped with a surjective homomorphism

$$= \mid \to$$

(1)

$$\mathcal{T}^{*\mid+\wedge}_{1,1} \simeq \mathrm{GL}(2,\mathbb{Z})^\wedge \xrightarrow{\text{epimorphism}} \mathcal{G}$$

,

which we will express by giving the three topological generators $\rho, \sigma, \tau' \in \mathcal{G}$, satisfying

$$(2) \quad \rho^6 = \sigma^4 = \tau'^2 = 1, \quad \begin{matrix} \rho^3 = \sigma^2 \\ (\text{denoted } -1) \end{matrix}, \quad \tau'\rho\tau'^{-1} = \rho^{-1}, \quad \tau'\sigma\tau'^{-1} = \sigma^{-1}.$$

We denote by $\mathcal{G}^+$ the closed subgroup generated by $\rho, \sigma$, which is thus the image of

$$\mathcal{T}^{+\wedge}_{1,1} \simeq \mathrm{SL}(2,\mathbb{Z})^\wedge,$$

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---

*Note marginale.*



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[scan 555]

thus $\{1,\tau'\}$ acts on these groups, and we can consider

$$\begin{cases} N_\rho = \{1,\tau'\}\,.\,Z_\rho, \\ N_\sigma = \{1,\tau'\}\,.\,Z_\sigma \end{cases} \qquad N \qquad \text{(products } \tfrac{1}{2} \text{ direct).} \tag{34}$$

We have canonical exact sequences

$$\begin{array}{ccccccccc} 1 & \longrightarrow & Z_\rho|L_\rho & \longrightarrow & N_\rho & \xrightarrow{\varepsilon_\rho} & \{\pm 1\} & \longrightarrow & 1 \\ 1 & \longrightarrow & Z_\sigma & \longrightarrow & N_\sigma & \xrightarrow{\varepsilon_\sigma} & \{\pm 1\} & \longrightarrow & 1 \end{array} \quad , \tag{35}$$

and a commutative diagram of canonical homomorphisms

$$\begin{array}{ccccc} & & \mathbb{D}_\rho & \longrightarrow & N_\rho \\ & \nearrow & & & \searrow^{i_\rho} \\ \{1,\tau'\}\times\{\pm 1\} & & & & G_{\rho,\sigma} \\ & \searrow & & & \nearrow_{i_\sigma} \\ & & \mathbb{D}_\sigma & \longrightarrow & N_\sigma \end{array} \tag{36}$$

where

$$\begin{cases} \mathbb{D}_\rho \simeq \mathbb{D}_6 \subset \mathrm{Gl}(2,\mathbb{Z}) \quad \text{generated by } \rho,\tau' \\ \mathbb{D}_\sigma \simeq \mathbb{D}_4 \subset \mathrm{Gl}(2,\mathbb{Z}) \quad \text{_____} \ \sigma,\tau' \\ \{1,\tau'\}\times\{\pm 1\} = \mathbb{D}_\rho \cap \mathbb{D}_\sigma \quad \text{in } \mathrm{Gl}(2,\mathbb{Z}), \end{cases} \tag{37}$$

hence the homomorphisms $\mathbb{D}_\rho \longrightarrow N_\rho$

$$\begin{cases} \mathbb{D}_\rho = \{1,\tau'\}\,.\,L_\rho \quad \text{(at least if } \rho \text{ of order 6 in } \mathcal{G}) \\ \mathbb{D}_\sigma = \{1,\tau'\}\,.\,L_\sigma \quad (\text{_____} \ \sigma \text{ of order 4 } \text{_____}), \end{cases} \tag{38}$$

and the homomorphisms

$$\mathbb{D}_\rho \longrightarrow N_\rho, \qquad \mathbb{D}_\sigma \longrightarrow N_\sigma$$

are obvious. The homomorphisms $i_\rho, i_\sigma$ too, of course.

Let us recall that

$$\begin{cases} i'_\rho i_\rho \text{ injective} \iff \tau' \notin Z_\rho \iff \rho^2 \neq 1 \iff \rho \neq -1 \\ \qquad\qquad \iff \mathcal{G}^+ \text{ is isomorphic|isomorphic to a quotient of } \mathbb{Z}/4\mathbb{Z}. \\ i_\sigma \text{ injective} \iff \tau' \notin Z_\sigma \iff \sigma^2 \neq 1 \quad \iff \sigma \neq \pm 1 \end{cases} \tag{39}$$

[p. 554]

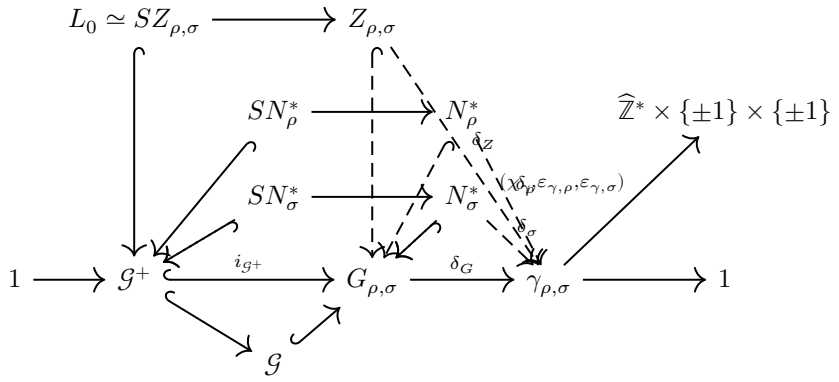


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[scan 557]

N.B. The notations $\varepsilon_\gamma, \varepsilon'_\gamma : \gamma_{\rho,\sigma} \longrightarrow \{\pm 1\}$ can lead to confusion, the first referring|pertaining to the action of a $u$ on $L_\sigma$ (not on $L_\rho$). A slightly heavier notation, but not leading to confusion, would be $\varepsilon_{\gamma,\rho}, \varepsilon_{\gamma,\sigma}$ — or even just $\varepsilon_\rho, \varepsilon_\sigma$, the confusion with the homomorphisms of this name $N'_\rho|N^*_\rho \longrightarrow \{\pm 1\}$, $N_\sigma \longrightarrow \{\pm 1\}$ being harmless when $\rho^2 \neq 1, \sigma^2 \neq 1$, because of the compatibilities (41) (43).

We have thus constructed, with the aid of|of the (1) i.e. of $\mathcal{G}$ equipped with $\rho, \sigma, \tau'$, groups $\mathcal{G}^+$, $G_{\rho,\sigma}, \gamma_{\rho,\sigma}, Z_{\rho,\sigma}, N'_\rho, N'_\sigma$, fitting into a commutative diagram of groups




(45)

where we have set

$$\begin{cases} SZ_{\rho,\sigma} = i_{L_0,Z}(L_0) = \mathrm{Ker}(\delta_Z : Z_{\rho,\sigma} \longrightarrow \gamma_{\rho,\sigma}) \\ SN'_\rho = \mathrm{Ker}(\delta_\rho : N^*_\rho \longrightarrow \gamma_{\rho,\sigma}) = \mathrm{Cent}_{\mathcal{G}^+}(\rho) \\ SN'_\sigma = \mathrm{Ker}(\delta_\sigma : N^*_\sigma \longrightarrow \gamma_{\rho,\sigma}) = \mathrm{Cent}_{\mathcal{G}^+}(\sigma) \end{cases} \tag{46}$$

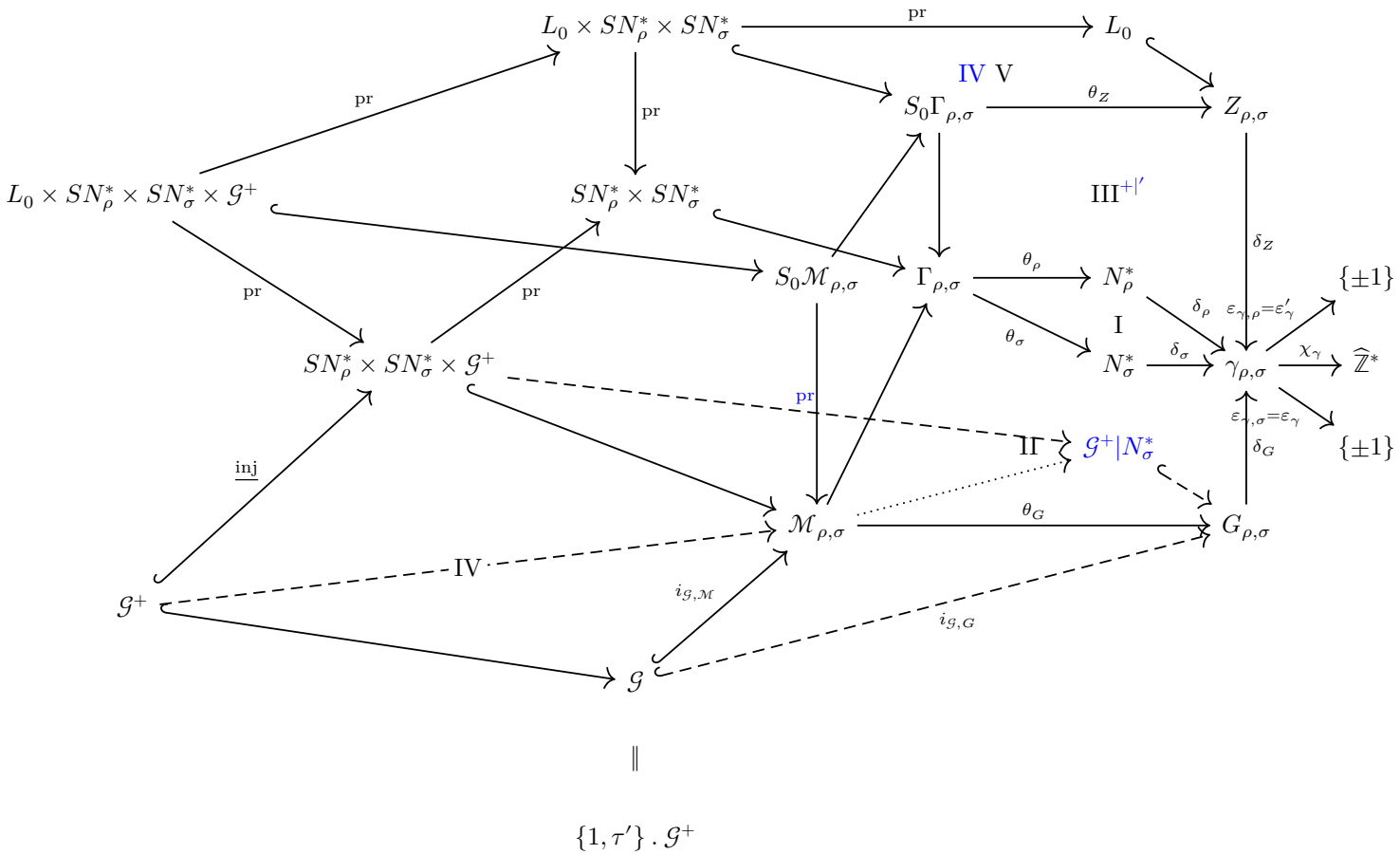
*Note marginale.* N.B. We can thus also denote them by $SZ_\rho, SZ_\sigma$ instead of the $SN^*_\rho, SN^*_\sigma$.

[p. 556]

[scan 561]

[p. 584]

[Diagramme continué sur la page en paysage; 140-3-561-D1]




(62)

$\|$

$\{1, \tau'\} \cdot \mathcal{G}^+$

We have summarized the mainsome of the relations between the groups introduced so far in the commutative diagram (62), which I comment on briefly.

a) All the arrows are either injections ($\hookrightarrow$) or surjections ($\longrightarrow$), with the exception perhaps of the three arrows issuing from $\gamma_{\rho,\sigma}$.

b) The ten squares (six faces of a cube, and fourthree|the squares marked $\mathrm{I}, \mathrm{II}, \mathrm{III}^{+|'}, \mathrm{IV}, \mathrm{V}$) are cartesian.

c) The five arrows marked pr are projections, canonical from product groups onto partial products. The arrow marked inj

$$\mathcal{G}^+ \longrightarrow SN_\rho^* \times SN_\sigma^* \times \mathcal{G}^+$$

is the canonical injection.

d) In order not to destroy the commutativity of the diagram, we have omitted to show in it the arrows (of inclusion, in particular)

$$Z_{\rho,\sigma} \longrightarrow G_{\rho,\sigma}, \qquad N_\rho^* \longrightarrow G_{\rho,\sigma}, \qquad N_\sigma^* \longrightarrow G_{\rho,\sigma}$$

which appeared in (45).

[p. 560]


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[scan 606]

To have The relation (99), it suffices to have is written

$$\begin{cases} \det\sigma a \equiv u \quad (\mathrm{SL}(2,\widehat{\mathbb{Z}})) \quad \text{i.e.} \quad \det\sigma a = \det u \quad \text{i.e. (98)} \\ \sigma a \equiv u \quad (42) \end{cases} \tag{102}$$

i.e. and this last congruence is written

$$\begin{gathered} c \equiv 0 \quad (42) \qquad d \equiv 1 \quad (42) \\ (-d \equiv \mu \quad (42) \text{ is a consequence of it for } \mu \equiv 1 \equiv -1 \quad (2)) . \end{gathered} \tag{103}$$

N.B. It is not possible to have the [illeg. 3] $\mu \equiv -1$ (4), for congruences mod 4 ($\sigma a \equiv u$ (4), for we would have $\mu \equiv -d \equiv -1$ (4), which is contrary to the hypothesis (95) $\mu \equiv 1$ (4).

Thus (102) is too strong. Nevertheless, we must (to have (99)) have

$$\sigma a \equiv u \quad (2) \qquad \text{i.e.} \tag{102 bis}$$

$$c \equiv 0 \quad (2), \qquad d \equiv 1 \quad (2) \tag{103 bis}$$

(the relation $-d \equiv \mu$ (2) is a consequence of it, for $\mu \equiv 1 \equiv -1$ (2)). This

Returning to the relation (99), we find this:

Proposition Lemma. Let $c, d \in \widehat{\mathbb{Z}}^*$ satisfy

$$\mu \overset{\text{dfn}}{=} -(c^2 + cd + d^2) \in \widehat{\mathbb{Z}}^* \tag{104}$$

---

*Note marginale.* N.B. We will necessarily have $\mu \equiv -1$ (3)

$$\mu \equiv 1 \quad (4) \qquad (\text{i.e. } c^2 + cd + d^2 \equiv -1 \quad (4)). \tag{105}$$

$$c \equiv 0 \quad (2), \qquad d \equiv 1 \quad (2) \tag{106}$$

Let a c|a= d

$$a = \tau'(c\rho + d) = \begin{pmatrix} -c & c+d \\ d & c \end{pmatrix} \in \mathrm{GL}(2,\widehat{\mathbb{Z}}) \tag{107}$$

(matrix with determinant $\mu$). Then we have a congruence

$$a\tau'(a)^{-1} \equiv \varepsilon \qquad \mod \widehat{\Pi}_0 \subset \mathrm{GL}(2,\widehat{\mathbb{Z}}) \tag{108}$$

and the sign $\varepsilon \in \{\pm 1\}$ does not depend on the choices made for $c, d$.

[scan 609]

[p. 608]

$$\mathcal{E} \simeq \{1, x\} \,.\, \mathcal{E}'' \qquad \left(\tfrac{1}{2} \text{ direct}\right) \tag{110}$$

Let us note that if $x$ operated trivially on $\mathcal{E}''$, i.e. if $\mathcal{E}$ were isomorphic to the product, then for an element $u$ of $\mathcal{E}''$ of order 4, $x.u$ would be of order 4|8, which is absurd (third proof of the fact that $\mathcal{E}/\mathcal{E}'' \simeq \mu$ operates nontrivially on $\mathcal{E}''$). Now from the mere fact that the operation of $\{1, x\}$ on $\mathcal{E}''$ is nontrivial., it is known, and the structure of $\mathcal{E}$ is determined – in fact we have

$$\mathcal{E} \simeq \mathbb{D}_4 \tag{111|110}$$

There is a preferred choice of $x$, by taking for $x$ the element

$$\begin{cases} \tau_{\mathcal{E}} \overset{\text{dfn}}{=} \tau_{S_0\Gamma} \bmod S_0^{\Pi}\Gamma^0, & \text{where of course} \\ \tau_{S_0\Gamma} = (\tau, \tau', \tau), \end{cases} \tag{112}$$

so that we have a canonical representation

$$\mathcal{E} \simeq \{1, \tau_{\mathcal{E}}\} \,.\, \mathcal{E}'' \qquad \text{product } \tfrac{1}{2} \text{ direct} \tag{113}$$

$\mathcal{E}'' \simeq \mathbb{Z}/4\mathbb{Z}$
$\tau_{\mathcal{E}}(u) = u^{-1}$
for $u \in \mathcal{E}''$.

The homomorphism $\mathcal{E} \to \mu \times \mu$ is recovered from its restrictions to the two factors

$$\begin{aligned} &\mu \simeq \{1, \tau_{\mathcal{E}}\} \longrightarrow \mu \times \mu \qquad \tau_{\mathcal{E}} \longmapsto (-1, -1) \quad \text{(diagonal homomorphism)} \\ &\mathcal{E}'' \longrightarrow \mu \times \{1\} \hookrightarrow \mu \times \mu \end{aligned} \tag{114}$$

the unique nontrivial homomorphism $\mathcal{E}'' \to \mu$ deduced from the isomorphism

$$\mathcal{E}''_2 \overset{\text{dfn}}{=} \mathcal{E}''/2\mathcal{E}'' | \mathcal{E}''/_2\mathcal{E}'' \xrightarrow{\sim} \mu$$

[scan 619]

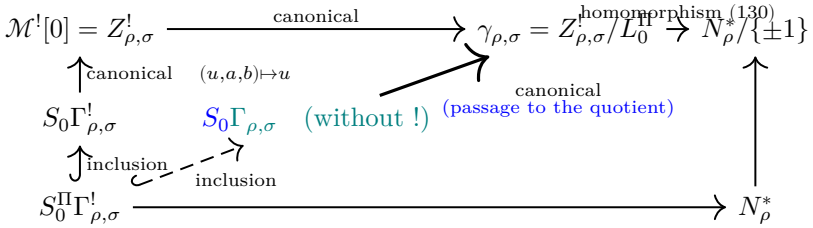
[p. 613]

Let us return to diagram (130), I say that the homomorphism

$$\begin{array}{ccc} \gamma_{\rho,\sigma} & \longrightarrow & N_\rho^*/\{\pm 1\} \\ \| & & \\ \Gamma_{\mathbb{Q}}^{\sim} & & \end{array}$$

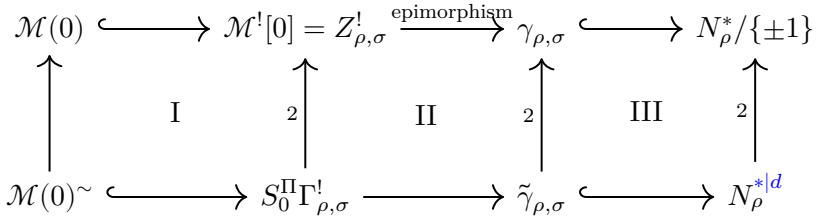
does not lift to a homomorphism from $\gamma_{\rho,\sigma}$ to $N_\rho^*$ — in other words, that the extension of $\gamma_{\rho,\sigma}$ by $\{\pm 1\} = \mu$ which it defines is not trivial. To identify this extension, let us consider the diagram N.B.

(144)




which shows that we have a diagram of cartesian squares

(145) (146)|(144)




where $\mathcal{M}(0)^\sim$ is defined|defined as the extension of $\mathcal{M}(0)$ by $\mu$ induced by the extension $S_0^\Pi\Gamma^!_{\rho,\sigma}$ of $\mathcal{M}^![0]$, which is thus identified also with the inverse|inverse image of

[scan 621]

[p. 614]

So we have an exact sequence

(149)

$$\begin{array}{ccccccccc} 1 & \to & \mu & \longrightarrow & Z_\rho^{*!} & \longrightarrow & \gamma_{\rho,\sigma} & \to & 1 \\ & & & & & & \wr\vert & & \\ & & & & (\simeq \tilde{\gamma}_{\rho,\sigma}) & & \tilde{\Gamma}_{\mathbb{Q}} & & \end{array}$$

which makes $Z_\rho^{*!}$ a $\mu$-extension of $\gamma_{\rho,\sigma}$, hence a canonical homomorphism

$$\gamma_{\rho,\sigma} \simeq Z_\rho^{*!}/\mu \longrightarrow N_\rho^*/\mu$$

which, of course, is none other than the first homomorphism (130).

9°) Recapitulation on the universal case.

Finally, I have the impression that I have again made long detours and contortions, to finally get lost in the sands, and lose the thread a little. So I am going to recapitulate what seems essential to me.

ⓐ

$$(1) \qquad Z_{\rho,\sigma} \simeq \mathcal{M}(0).L_0^{\mathcal{G}} = \mathcal{M}[0] \ [\textit{illeg. 1}]$$

(operating on $\widehat{\mathcal{G}}^+$ by operation induced by the adjoint)

we distinguish inside it a subgroup of index 2

$$(2) \qquad Z_{\rho,\sigma}^{!} \simeq \mathcal{M}(0).L_0^{\Pi}.$$

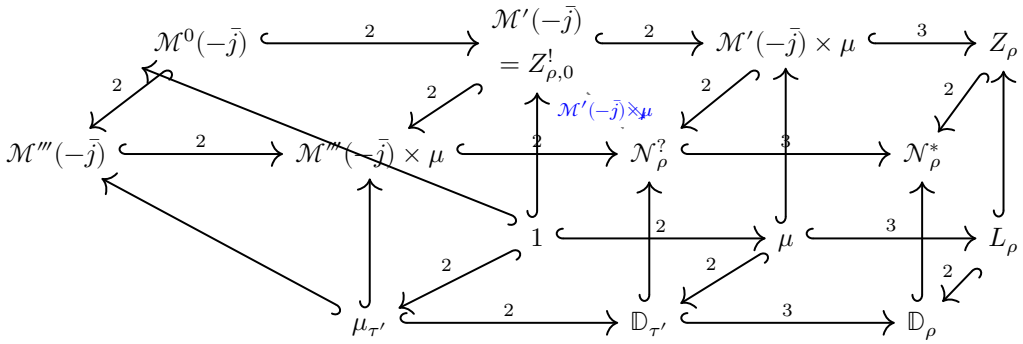
[scan 637]

[p. 622]

$$\begin{cases} \mathcal{M}''' \simeq \mathcal{M}'''(-\bar{j}) \,.\, \mathcal{G}^{+\wedge} \\ \mathcal{N}_\rho^* \simeq \mathcal{M}'''(-\bar{j}) \,.\, L_\rho. \end{cases} \tag{36}$$

Summary diagram of the subgroups of $\mathcal{N}_\rho^*$ obtained so far:

(37)




,

where we are here interested above all in the "upper face", containing|concerning subgroups of finite index in $\mathcal{N}_\rho^*$, intermediate between it and $\mathcal{M}_0(-\bar{j})|\mathcal{M}^0(-\bar{j})$ (of index 24 in $\mathcal{N}_\rho^*$). I am going to examine the question of invariance of these subgroups in one Another, and the structure of the quotient groups.

I am going to redo a diagram which really|better shows the groups|relationships relations between the groups in question. 636

[scan 653]

[p. 629]

2°) $Z_{\rho,\sigma},\ \gamma_{\rho,\sigma}$.

Let

$$Z_{\rho,\sigma} \subset \mathrm{Aut}(\mathcal{G}^+) \times \widehat{\mathbb{Z}}^* \times \mu \times \mu$$
$$\| \text{ dfn}$$
$$(4) \qquad \left\{ (u,\mu,\varepsilon_2,\varepsilon_3) \;\middle|\; \begin{array}{l} u \text{ normalizes } \mathcal{G}^\natural \text{ and induces the identity in } \mathcal{G}^+/\mathcal{G}^\natural \\ a)\ \exists \alpha \in \mathcal{G}_\tau^{\natural|\sharp},\ \text{with } \varepsilon_{\mathcal{G}}(\alpha) = \varepsilon_3, \quad u(\rho) = \alpha(\rho) \\ b)\ \exists \beta \in \mathcal{G}_\tau^{\natural|\sharp},\ \text{with } \varepsilon_{\mathcal{G}}(\beta) = \varepsilon_2, \quad u(\sigma) = \beta(\sigma) \\ c)\ u(\varepsilon_0) = \varepsilon_0^\mu \\ d)\ u \text{ normalizes } \mathcal{G}^{\natural|\sharp} \text{ and induces the identity on } (\mathcal{G}^+/\mathcal{G}^{\natural|\sharp})/\operatorname{Im}\Pi \end{array} \right\}.$$

---

*Note marginale.* cf. page 635 continuation a weaker condition this condition.

---

*Note marginale.* The last condition added in d) (in principle), is a consequence of a)b) if $\varepsilon_2\varepsilon_3 = 1$ (in which case $u|(\mathcal{G}^+/\mathcal{G}^{\natural|\sharp})$ the identity induced by $\varepsilon[\tau]$, where $\varepsilon = \varepsilon_2 = \varepsilon_3 \in \{\pm 1\}$).

Proposition. $Z_{\rho,\sigma}$ is a subgroup of the product group.

Proof. The conditions c), d) define obviously a each subgroup. Given that

$$\tau(\sigma) = -\sigma|\varepsilon,$$

condition b) also means that

$$(5^*) \qquad b') \quad u(\sigma) \text{ is } \mathcal{G}^{\natural|\sharp}\text{-conjugate to : } \ \varepsilon_2\sigma^{\varepsilon_2}(\varepsilon_2\sigma = \varepsilon_2\sigma)$$

and in this form it is clear that condition c') defines a subgroup. I do not think that condition a) by itself defines a subgroup. But if $\mathfrak{g}_{(b)}|G_{(b)}\ \mathfrak{g}_{(a,b)}$ denotes the subgroup of the product defined by these conditions a)b), we have to show that $\mathfrak{g}|G_{(a,b,c)(a,b)}$ is a subgroup of the product $\mathfrak{g}|G_{(a,b)(b)}$ — which will prove the proposition. Let $\underline{u}, \underline{u}' \in \mathfrak{g}_{(b)}|G_{(b)}$, corresponding to $(u,\mu,\varepsilon_2,\varepsilon_3)$ and $(u',\mu',\varepsilon_2',\varepsilon_3')$, and to $\alpha,\alpha' \in \mathcal{G}_\tau^{\natural|\sharp}$, so that we have

[p. 652]

[scan 656]

Remarks a) For condition b) to be multiplicative, we do not need condition a) in all its strength, it suffices that $u$ in $\mathcal{G}^+/\mathcal{G}^{\natural|\sharp}$ normalize the image of $L_\sigma$, i.e. that we have

$$u(\sigma) \equiv \pm\sigma \quad (\mathcal{G}^{\natural|\sharp}).$$

b) If $\mathcal{G}$ is "big enough", the projection homomorphism is injective

(12)

$$Z_{\rho,\sigma} \hookrightarrow \mathrm{Aut}(\mathcal{G}^+)$$

(cf. page 578)

Examples in the universal case

a) $\mathcal{G}^{\natural|\sharp} = \mathcal{G}^+ = \mathcal{G}\mathrm{Sl}(2,\mathbb{Z})^\wedge$. Here we have the Condition|condition (4) d) is empty, conditions a), b) are equivalent respectively to the fact that $u(\rho)$ conjugate to $\rho^{\varepsilon_3}$, $u(\sigma)$ conjugate to $\sigma^{\varepsilon_2}$, i.e. comes back to definition (6) page 578, which gives

$$Z_{\rho,\sigma} = M(\omega|0).L_0^{\mathcal{G}} = M^{\mathcal{G}}[0].$$

$$(13) \qquad Z_{\rho,\sigma} = M(0).L_0^{\mathcal{G}} = M^{\mathcal{G}}[0] = M[0]$$

*Note marginale.* N.B. A priori we rather have $Z_{\rho,\sigma} \subset \widetilde{M}_{0,3}|M_{0,3}$, but

$$\begin{array}{ccc} M[0] & \xrightarrow{\sim} & \widetilde{M}_{0,3}[0] \\ \| & & \| \\ M(\omega|0).L_0^{\mathcal{G}} & \xrightarrow{\sim} & M_{0,3}(0).L_0^{\mathcal{G}_{0,3}} \end{array}$$

b) $\mathcal{G}^{\natural|\sharp} = \Pi = \Pi_0 \times \mu$. We now find

$$(14) \qquad Z_{\rho,\sigma} = M(0).L_0^{\Pi} = M^{\Pi}[0] = M^{!}[0]|M'^{!}[0]$$

(denoted $Z'_{\rho,\sigma}$ in V number° 8).

*Note marginale.* In this case, it is immediate that $Z_{\rho,\sigma}$ operates trivially even on $\mathcal{G}^+/\mathcal{G}_\tau^{\natural|\sharp}$, for this is so for the N.B. all the subgroups contained by a), b) of (4), $\mathcal{G}^\natural = M(0)_2.L_0^{\Pi}|\mathcal{G}^\sharp = M(\omega)_2.L_0^{\Pi}$.

*Note marginale.* $M^{!}[0]|M'^{!}[0] = M^{\Pi}[0]$.

*Note marginale.* make explicit again why (i.e. why) that $(\varepsilon_0, 1, 1, 1) \notin Z_{\rho,\sigma}$

c) $\mathcal{G}^{\natural|\sharp} = \Pi_0$. Conditions a) and b) of (4) have not changed with respect to|with respect to the beforepreceding case, since $\Pi = \Pi_0 \times \mu$ and $\mu$ is central, on the other hand the

[p. 655]

[scan 658]

except by restricting ourselves to some, $(u, \mu, \varepsilon_2, \varepsilon_3)$ with $\varepsilon_3 = 1$, since $\dot\rho^{-1}$ is not equal to $\dot\rho$ ! It is true that (in the universal case for example) that $u(\rho)$ is conjugate in $\mathcal{G}^+$ to $\rho^{\varepsilon_3}$, but in general it will not be so by an element (say) of $\Pi$.

We set

$$(18) \qquad L_0^{\natural|\sharp} = L_0 \cap \mathcal{G}^{\natural|\sharp}\{l \in L_0^{\mathcal{G}} \mid (\mathrm{int}(l), 1, 1, 1) \in Z_{\rho,\sigma}\}$$

$$\left(\text{where } L_0^{\mathcal{G}} \subset \mathcal{G}^+ \text{ is the set of the } \varepsilon_0^n,\ n \in \widehat{\mathbb{Z}}\right)$$

*Note marginale.* N.B. We have $L_0^{\natural|\sharp} \supset L_0 \cap \mathcal{G}^{\natural|\sharp}$, and the inclusion can be strict, cf. example c) above…where $L_0^{\natural|\sharp} = L_0^{\Pi}$, $L_0 \cap \mathcal{G}^{\natural|\sharp} = L_0^{\Pi_0}$. which is surprising.

and we have a canonical homomorphism

$$(19) \qquad L_0^{\natural|\sharp} \xrightarrow{\ i\ } Z_{\rho,\sigma},$$

$$l \longmapsto (\mathrm{int}_{\mathcal{G}^+}(l), 1, 1, 1)$$

injective iff we have

$$(20) \qquad L_0^{\natural|\sharp} \cap \mathrm{Centr}(\mathcal{G}^+) = \{1\}$$

the image invariant. We set

$$(21) \qquad \gamma_{\rho,\sigma} = Z_{\rho,\sigma}/L_0^{\natural|\sharp},$$

whence

$$(22) \qquad 1 \longrightarrow L_0^{\natural|\sharp} \longrightarrow Z_{\rho,\sigma} \xrightarrow{\delta_Z} \gamma_{\rho,\sigma} \longrightarrow 1,$$

$$(23) \qquad Z_{\rho,\sigma} \longrightarrow \gamma_{\rho,\sigma} \begin{cases} \xrightarrow{\varepsilon_{3\gamma}} \mu \\ \xrightarrow{\chi_\gamma} \widehat{\mathbb{Z}}^* \\ \xrightarrow{\varepsilon_{2\gamma}} \mu \end{cases}$$

) deduced from the

$$(u, \mu, \varepsilon_2, \varepsilon_3) \begin{cases} \mapsto \varepsilon_3 \\ \mapsto \mu \\ \mapsto \varepsilon_2. \end{cases}$$

Remarks We denote set,

$$(24) \qquad \begin{cases} \gamma'_{\rho,\sigma} = \mathrm{Ker}\,\varepsilon_{3\gamma}, & \gamma''_{\rho,\sigma} = \mathrm{Ker}\,\varepsilon_{2,\sigma}|\varepsilon_{2\gamma}, & \gamma'''_{\rho,\sigma} = \mathrm{Ker}\,\varepsilon_{2\gamma}\varepsilon_{3\gamma} \\ Z'_{\rho,\sigma} = \ldots & Z''_{\rho,\sigma} = \ldots & Z'''_{\rho,\sigma} = \ldots \end{cases}$$

(inverse images in $Z_{\rho,\sigma}$)

[p. 657]

[scan 659]

[p. 632]

We set

$$(24)\qquad \begin{cases} \gamma'_{\rho,\sigma} = \operatorname{Ker}\varepsilon_{3\gamma}, \\ \gamma''_{\rho,\sigma} = \operatorname{Ker}\varepsilon_{2\gamma}, \\ \gamma'''_{\rho,\sigma} = \operatorname{Ker}\varepsilon_{2\gamma}\varepsilon_{3\gamma},\ \gamma^{\circ}_{\rho,\sigma} = \gamma' \cap \gamma'' = \gamma'' \cap \gamma''' = \gamma''' \cap \gamma' \\ Z'_{\rho,\sigma}, Z''_{\rho,\sigma}, Z'''_{\rho,\sigma}, Z^{\circ}_{\rho,\sigma} \text{ are the inverse images of the } Z_{\rho}\gamma'_{\rho,\sigma} \text{ etc} \\ \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \text{in } Z_{\rho,\sigma} \end{cases}$$

These are thus subgroups of index 1, or 2 or four in $\gamma_{\rho,\sigma}$ resp. $Z_{\rho,\sigma}$ — the index 4 being possible only for $\gamma^{\circ}_{\rho,\sigma}, Z^{\circ}_{\rho,\sigma}$.

Remarks 1) We thus have

$$(25)\qquad Z^{\circ}_{\rho,\sigma} \simeq \left\{ (u,\mu) \in \operatorname{Aut}(\mathcal{G}^{+}) \times \widehat{\mathbb{Z}}^{*} \;\middle|\; \begin{array}{ll} \text{a)} & u(\rho) \text{ is } \mathcal{G}^{\natural|\sharp}\text{-conjugate to } \rho \\ \text{b)} & u(\sigma) \quad\underline{\qquad}\quad \sigma \\ \text{c)} & u(\varepsilon_0) = \varepsilon_0^{\mu} \end{array} \right\},$$

*Note marginale.* More generally,

$$Z'''_{\rho,\sigma} \simeq \left\{ (u,\mu,\varepsilon) \in \operatorname{Aut}(\mathcal{G}^{+}) \times \widehat{\mathbb{Z}}^{*} \times \mu \;\middle|\; \begin{array}{ll} \text{a)} & u(\rho) \text{ is } \varepsilon[\tau](\rho)\mathcal{G}^{\natural|\sharp}\text{-conjugate to } \varepsilon[\tau](\rho) \\ \text{b)} & u(\sigma) \text{ is } \mathcal{G}^{\natural|\sharp}\text{-conjugate to } \varepsilon[\tau](\sigma) \\ \text{c)} & u(\varepsilon_0) = \varepsilon_0^{\mu} \end{array} \right\},$$

(isomorphism of subgroups). The condition d) of (4) — stability of $\mathcal{G}^{\natural|\sharp}$ under $u$ — is a consequence here of a) b) c), for we will have

$$u(\rho) \equiv \rho \quad (\mathcal{G}^{\natural|\sharp})$$
$$u(\sigma) \equiv \sigma \quad (\mathcal{G}^{\natural|\sharp})$$

which implies

$$(25\text{bis})\qquad u(x) \equiv x \quad (\mathcal{G}^{\natural|\sharp}) \qquad \forall x \in \mathcal{G}^{+},$$

i.e. the stability of $\mathcal{G}^{\natural|\sharp}$ and a trivial action of $u$ on $\mathcal{G}/\mathcal{G}^{\natural|\sharp}$.

2) We also have the remarkable element

$$(26)\qquad \tau_Z \in Z_{\rho,\sigma}, \qquad \tau_Z = (\tau, -1, -1, -1)$$

*Note marginale.* That is one more reason not to require in (4) d) that $u$ operate trivially on $\mathcal{G}^{+}/\mathcal{G}^{\natural|\sharp}$, which would make me|we would make lose $\tau$ !

3°) $S_0G_{\rho,\sigma}, G_{\rho,\sigma}, S_0G^{\natural|\sharp}_{\rho,\sigma}, G^{\natural|\sharp}_{\rho,\sigma}$.

Proceeding as on page 577, we find

$$Z_{\rho,\sigma} \longrightarrow \operatorname{Aut}(\mathcal{G}^{+})$$

hence 1/2 direct products

$$(27)\qquad S_0G_{\rho,\sigma} \overset{\text{dfn}}{=} Z_{\rho,\sigma}\,.\,\mathcal{G}^{+}, \supset S_0G^{\natural|\sharp}_{\rho,\sigma} \overset{\text{dfn}}{=} Z_{\rho,\sigma}\,.\,\mathcal{G}^{\natural|\sharp}$$

and a homomorphism

$$(28)\quad L_0^{\natural|\sharp} \xhookrightarrow{i_{L_0,S_0G}|i_{L_0,G}} S_0G^{\natural|\sharp}_{\rho,\sigma} \qquad \subset \qquad S_0G_{\rho,\sigma}$$

$$l \longmapsto \iota(l)|i(l)\,i_{L_0,Z}(l)\,.\,l^{-1}$$

[p. 658]

[scan 662]

central in $Z_0^{\natural|\sharp}$. Let us consider the map

$$Z_0^{\natural|\sharp} \longrightarrow S_0 G_{\rho,\sigma}^{\natural|\sharp} = Z_{\rho,\sigma} \cdot \mathcal{G}^{\natural|\sharp}$$

$$\begin{array}{ccccc} f \mapsto f^{-1} & \cdot \underline{u}(f) & = \underline{u}(f) \cdot f^{-1} & \text{where } \underline{u}(f) = (\mathrm{int}(f), 1, 1, 1) \in Z_{\rho,\sigma} \\ \cap & \cap & & \cap \\ \mathcal{G}^{\natural|\sharp} & Z_{\rho,\sigma} & & G_{\rho,\sigma}^{\natural|\sharp} \end{array} \tag{42}$$

This map is a homomorphism of groups, whose|definitionit inverse image of $Z_0|L_0$ $L_0$ induces|injective by projection|passage to the quotient $L_0^{\natural|\sharp}$ the image induces on $L_0^{\natural|\sharp} \subset Z_0^{\natural|\sharp}$ the canonical homomorphism $i_{L_0,SG}|i_{L_0,S_0G}$ (28), and $L_0^{\natural|\sharp} \subset Z_0^{\natural|\sharp}$ is the inverse image of

$$i_{L_0,SG}|i_{L_0,S_0G}(L_0^{\natural|\sharp}) = \mathrm{Ker}(S_0 G_{\rho,\sigma}^{\natural|\sharp} \longrightarrow G_{\rho,\sigma}^{\natural|\sharp}).$$

We thus find an injective homomorphism

$$Z_0^{\natural|\sharp}/L_0^{\natural|\sharp} \hookrightarrow G_{\rho,\sigma}^{\natural|\sharp} \tag{43}$$

whose image is the kernel of (38), i.e. we have a canonical exact sequence

$$\begin{array}{ccccccccc} 1 & \to & L_0^{\natural|\sharp} & \to & Z_0^{\natural|\sharp} & \to & G_{\rho,\sigma}^{\natural|\sharp} & \longrightarrow & \widetilde{G}_{\rho,\sigma}^{\natural|\sharp} \longrightarrow 1 \\ & & & & & & \cap & & \\ & & & & & & \mathrm{Aut}(\mathcal{G}^+) \times \widehat{\mathbb{Z}}^* \times \mu \times \mu & & \end{array} \tag{44}$$

where $\widetilde{G}_{\rho,\sigma}^{\natural|\sharp}$ consists of the $(u, \mu, \varepsilon_2, \varepsilon_3)$ satisfying the conditions (39).

Remark. This proposition brings out to what extent the [illeg. 1] construction made here of $G_{\rho,\sigma}^{\natural|\sharp}$ is finer than the one where we would define|defined it simply as a group of automorphisms of $\mathcal{G}^+$ (we even have|of them need of the $\mu, \varepsilon, \varepsilon'$ into the bargain…). [p. 661]

[scan 20]

with the aid by giving oneself a system

$$(35)\qquad \begin{array}{c}\underline{\mathcal{G}} \supset \underline{M}\\ \text{Group over } S_0,\\ \text{with } \underline{M} \text{ as}\\ \text{normal subgroup}\end{array} \qquad \text{, and } \Phi\colon \mathcal{G}_{S_0} \longrightarrow \underline{\mathcal{G}}$$

such that (34) is reconstituted from (35) by taking $\underline{M}$, $\mathcal{G}_{S_0}$ in (35), $M_{S_0}$ [illeg. 1]= the inverse image of $\underline{M} \subset \underline{\mathcal{G}}$ under $\Phi$, $\varphi$ induced by $\Phi$, and $\Theta$ the operation deduced from $\Phi$. Moreover, $\Phi$ induces an isomorphism.

$$\underbrace{\mathcal{G}_{S_0}/M_{S_0}}_{H_{S_0}} \xrightarrow{\sim} \underline{\mathcal{G}}/\underline{M}$$

– and we have here an "axiomatic" description of the system (35) in terms of (34). The exact sequence of schemesGroups over $S|S_0$

$$(36)\qquad 1 \longrightarrow \underline{M} \longrightarrow \underline{\mathcal{G}} \longrightarrow \underline{\mathcal{G}}/\underline{M} \longrightarrow 1$$
$$\wr$$
$$H_{S_0}.$$

in fact, over $\forall$ affine base scheme $S$ (more generally, $S$ such that $H^1(S, M \otimes_{\mathbb{Z}} \underline{\mathcal{O}}_S) = 0$) some exact sequences

---

*Note marginale.* The hypothesis made about $H^1$ being zero|zero is entirely useless, for the $\underline{M}$-torsors over $S_0$ defined by the inverse images of the $h(\in H)$ are obviously trivial

$$(37)\qquad \begin{array}{ccccccccc} 1 & \longrightarrow & \underline{M}(S) & \longrightarrow & \underline{\mathcal{G}}(A)|(/A) & \longrightarrow & H_{S_0}(S) & \longrightarrow & 1\\ & & \wr & & & & \wr & & \\ & & M \otimes_{\mathbb{Z}} A & & & & H \text{ if } S \text{ connected} & & \end{array}$$

if $S$ affine with ring $A$

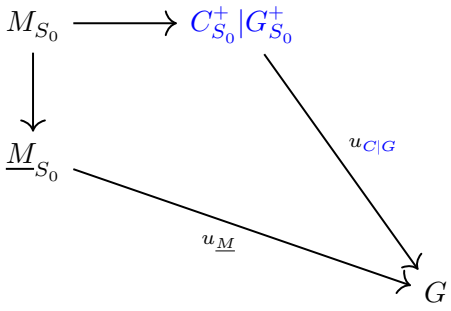
[scan 44]

define homomorphisms

$$\underline{M}_{S_0} \xrightarrow{u_{\underline{M}}} G^\sharp|G^\natural \ , \qquad C_{S_0}|G_{S_0} \xrightarrow{u_{C|G}} G^\sharp|G^\natural \quad \text{i.e.} \quad C|G \to G^\sharp(S_0)|G^\natural(S_0) \tag{40}$$

satisfying the following two conditions:

a) Commutativity of the diagram




, (41)

and b)

$$\begin{gathered} u_{\underline{M}} \text{ is compatible with the actions} \\ \text{of } C^+_{S_0}|G^+_{S_0} \text{ on } \underline{M}_{S_0} \text{ (via its action on } M_{S_0}) \\ \text{and on } G \text{ (via } u_{C|G}). \end{gathered} \tag{42}$$

In the present case, we will have $G = \underline{C}|\underline{G}$

$$u_{\underline{M}}(\lambda) = \lambda\, i(\mu.\lambda) \tag{43}$$

where

$$i : \quad \underline{M} \hookrightarrow \underline{C}^+|\underline{G}^+$$

is the inclusion, and

$$u_{C|G} : \text{defined by } C^+_{S_0}|G^+_{S_0} \longrightarrow G \qquad \text{defined by (33)}, \tag{44}$$

We must therefore first check that (33), does define a homomorphism

$$C_{S_0}|G_{S_0} \to G$$

, or what amounts to the same,

$$C|G \to G(S_0)$$

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*Note marginale.* 43


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[scan 66]

that $M_0$ the subgroup under consideration of $\Pi_0 \subset \mathrm{Sl}(2,\mathbb{Z})$, of index 3 in $\Pi_0$, is the inverse image of this preceding subgroup in $M$ of $\mathrm{Sl}(2,\mathbb{Z}/12\mathbb{Z})$.

Let us sum up the results obtained:

Proposition Let us consider in $\mathcal{G}_0^+ = \mathrm{Sl}(2,\mathbb{Z})$ the tower of invariant subgroups (even in $\mathcal{G}_0$)

(114)

$$\mathcal{G}_0^+ \xleftarrow{6} \Pi = \Pi[2] \xleftarrow{2} \Pi_0 \xleftarrow{3} \Pi_{00}\ (\supset [\Pi_0,\Pi_0])$$

$\Pi = \Pi[2]$: $\overset{\text{dfn}}{=}$ $\{u \in \mathcal{G}_0^+ \mid u \equiv 1\ (2)\}$ $=$ $\mathfrak{sl}(2,\mathbb{F}_2) \simeq \{u \in \mathrm{M}_2(\mathbb{F}_2) \mid \mathrm{Tr}\, u = 0\}$ (i)|(1)

$\Pi_0$: $=$ Subgroup generated by the $-\ell_i = \lambda_i$

$\Pi_{00}$ $(\supset [\Pi_0,\Pi_0])$: $=$ Subgroup invariant generated by by the $\lambda_i$, $[\lambda_0,\lambda_1]$ and $\lambda_1\lambda_0^{-1} = \ell_1\ell_0^{-1}$

Let us consider on the other hand the quotient group

(115)
$$\begin{array}{ccc} \mathrm{Sl}(2,\mathbb{Z}/12\mathbb{Z}) & \simeq & \mathcal{G}_0^+/\underbrace{\Pi[12]}_{\simeq\{u\in\mathcal{G}_0^+ \mid u\equiv 1\ (12)\}} \\ \simeq & & \\ \mathrm{Sl}(2,\mathbb{Z}/4\mathbb{Z}) \times \mathrm{Sl}(2,\mathbb{Z}/3\mathbb{Z}), & & \end{array}$$

and the tower of the following subgroups

(116)
$$\left\{\begin{array}{lll} & H = \mathrm{Sl}(2,\mathbb{Z}/4\mathbb{Z}) \times \mathrm{Sl}(2,\mathbb{Z}/3\mathbb{Z}) & \text{order } 48\times 24 \\ 6 \cup & \cup & \\ & \Pi^\bullet = \mathfrak{sl}(2,\mathbb{F}_2) \times \mathrm{Sl}(2,\mathbb{Z}/3\mathbb{Z}) & \text{order } 8\times 24 \\ 2 \cup & \cup & \\ & \Pi_0^\bullet = \mathfrak{sl}(2,\mathbb{F}_2)' \times \mathrm{Sl}(2,\mathbb{Z}/3\mathbb{Z}) & \text{order } 4\times 24 \\ 3 \cup & \cup & \\ & \Pi_{00}^\bullet = \mathfrak{sl}(2,\mathbb{F}_2)' \times \underbrace{\mathrm{Syl}_2}_{\simeq[illeg.\ 1]D_4} & \text{order } 4\times 8, \end{array}\right.$$

---

*Note marginale.* N.B. $(\mathcal{G}_0^+)_{\mathrm{ab}} \simeq \mathbb{Z}/12\mathbb{Z}$ is identified with a (115)|(116) Sl(2) quotient of $H/\Pi_{00}^\bullet$ by a subgroup (116) of order 3 — that is again the subgroup generated by the element $(-\dot\rho, 1)$ of

$$\underbrace{\mathrm{Sl}(2,\mathbb{Z}/4\mathbb{Z})/\mathfrak{sl}(2,\mathbb{F}_2)'}_{\mathcal{G}_0^+/\Pi_0} \times \mathrm{Sl}(2,\mathbb{Z}/3\mathbb{Z})/\mathrm{Syl}_2.$$

Where we have

(117)
$$\left\{\begin{array}{ll} \text{a)} & \mathfrak{sl}(2,\mathbb{F}_2) \simeq \\ & \mathrm{Ker}(\mathrm{Sl}(2,\mathbb{Z}/4\mathbb{Z}) \longrightarrow \mathrm{Sl}(2,\mathbb{Z}/2\mathbb{Z})) \\ & \qquad\qquad \text{normal subgroup } \simeq \\ & \qquad\qquad \mathbb{F}_3|\mathbb{F}_2\ \mathbb{F}_2\times\mathbb{F}_2\ ;\ \text{whose three elements } \neq 0 \\ & \qquad\qquad \text{are } \dot\lambda_0, \dot\lambda_1, \dot\lambda_\infty \ \text{ (images} \\ \text{b)} & \mathfrak{sl}(2,\mathbb{F}_2)' \subset \mathfrak{sl}(2,\mathbb{F}_2), \quad \text{in } \mathrm{Sl}(2,\mathbb{Z}/4\mathbb{Z})) \text{ — it is} \\ & \qquad\qquad \text{a complement to its} \\ & \qquad\qquad \text{Lie subalgebra } \mu \simeq \mathbb{F}_2 \text{ of matrices} \\ & \qquad\qquad \text{constant|scalar} \\ \text{c)} & \mathrm{Syl}_2 \subset \mathrm{Sl}(2,\mathbb{Z}/3\mathbb{Z}), \\ & \quad \| \\ & \mathrm{Ker}(\mathrm{Sl}(2,\mathbb{Z}/3\mathbb{Z}) \to \mathcal{G}_3^+) \\ & \text{Sylow subgroup, of order 8} \\ & \text{(which is normal)} \end{array}\right.$$

[scan 113]

[p. 704]

substituting these values of $x, y$ into it, a relation linking $a_0, a_1, b, \nu$:

$$\nu(-4a_1 + 6b - 2) = -2\nu a_0 \qquad \text{i.e.}$$

$$2\nu(a_0 - 2a_1 + 3b - 1) = 0$$

which is a consequence of (263)! Thus we find: that for a splitting

Proposition For every splitting of the extension $\underline{\mathfrak{S}}^+/\mu_{\underline{\mathfrak{S}}}$ of $(\underline{\mathfrak{S}}_3)_{S_0}$ by $\underline{M}$ (over an arbitrary base), the commutant in $\underline{G}_{\rho,\sigma}/\underline{\mu}_G$ of the section subgroup is again|a a section subgroup of the projection the extension $\underline{G}^{\natural|\sharp}_{\rho,\sigma}$ of $\mathbb{O}^*_{12}$ by $\underline{M}$. [The lifting

$$(264) \qquad u_{\rho',\sigma'} : \quad \mathbb{O}^*_{12} \longrightarrow \underline{G}^{\natural|\sharp}_{\rho,\sigma}$$

which is associated with $\rho', \sigma'$ given by (257)

$$\rho' = \zeta\rho, \quad \sigma' = \eta\sigma \qquad \begin{matrix} \zeta = a_0\lambda_0 + a_1\lambda_1, \quad \eta = b\delta = b(\lambda_1 - \lambda_0) \\ a_0, a_1, b \in \Gamma\mathbb{G}_a \end{matrix} ,$$

is given by

$$(265) \qquad u_{\rho',\sigma'}(\underline{\mu}) = \lambda u_{\underline{\mu}}$$

where $u_{\underline{\mu}}$ is defined in (150,151), and where $\lambda = x\lambda_0 + y\lambda_1$, with $x, y \in \Gamma\mathbb{G}_a$ described by 264, where where $\nu = \nu(\underline{\mu})$).

Corollary The canonical morphism

$$(266) \qquad \underline{\text{Scindages}}\left(\underline{G}_{\rho,\sigma}/\underline{\mu}_G \longrightarrow |\text{overover } \mathbb{O}^*_{12} \times \underline{\mathfrak{S}}_3\right) \to \underline{\text{Scindages}}\left(\underline{\mathfrak{S}}^+ \text{ over } \underline{\mathfrak{S}}_3\right)$$

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[scan 271]

[p. 783]

inverse images under $u_{\mathbb{S}|\mathcal{S}}$ and under $u_{\mathbb{D}|D^*}$ of $\mathbb{S}|\mathcal{S}^{\sim}_{x'}$ and of $\mathbb{D}|D^{*\sim}_{x'}$, that is|that is $\mathbb{S}|\mathcal{S}'^{\sim}_{x}$, $\mathbb{D}|D^{*\prime\sim}_{x}$, are equipped with a correlation, which I denote by $h'_x$. This posited, the existence of an isomorphism $F(u) \simeq G(u)$ canonical isomorphisms of $T_x$-torsors

$$(121)\quad \begin{aligned} F(u) &\simeq \mathrm{Isom}_{\mathbb{S}|\mathcal{S}_x}(\mathbb{S}|\mathcal{S}^{\sim}_{x}, \mathbb{S}|\mathcal{S}'^{\sim}_{x}) \\ G(u) &\simeq \mathrm{Isom}_{\mathbb{D}|D^*_x}(\mathbb{D}|D^{*\sim}_{x}, \mathbb{D}|D^{*\prime\sim}_{x}) \end{aligned}$$

Now by the diagram of equivalences (117) we do have isomorphisms of $T_x$-torsors

(122)

$$\begin{array}{ccc} & \mathrm{Isom}((\mathbb{S}|\mathcal{S}^{\sim}_{x}, \mathbb{D}|D^{*\sim}_{x}, h_x), & \\ & (\mathbb{S}|\mathcal{S}'^{\sim}_{x}, \mathbb{D}|D^{*\prime\sim}_{x}, h'_x)) & \\ \overset{\sim}{\swarrow} & & \overset{\sim}{\searrow} \\ F(u) = & & G(u) = \\ \mathrm{Isom}(\mathbb{S}|\mathcal{S}^{\sim}_{x}, \mathbb{S}|\mathcal{S}'^{\sim}_{x}]|) & & \mathrm{Isom}(\mathbb{D}|D^{*\sim}_{x}, \mathbb{D}|D^{*\prime\sim}_{x}) \end{array} ,$$

whence the canonical bijection sought (120) — of the which gives in particular (96).

The The canonical bijection (120) has the following ± obvious property, which characterizes it:

(123) We Let $u_{\mathbb{S}|\mathcal{S}^{\sim}}$ and $u_{\mathbb{D}|D^{*\sim}}$ be extensionsliftings of $u$ $u_{\mathbb{S}}$associated $\mathbb{S}|\mathcal{S}^{\sim}_{x} \simeq \mathbb{S}|\mathcal{S}^{\sim}_{x'}$, $\mathbb{D}|D^{*\sim}_{x} \longrightarrow \mathbb{D}|D^{*\sim}_{x'}$, associated by (120), let $A \in \mathcal{A}_x$, $A' \in \mathcal{A}'_x$ its image under $u_{\mathbb{S}|\mathcal{S}}$, let $\alpha$ be a section of $\mathbb{S}|\mathcal{S}^{\sim}_{x}$ over $A$, $\beta$ the section of $\mathbb{D}|D^{*\sim}_{x}$ over $\mathrm{Ep}|E_p(A)$ associated|associated by the correlation $h_x$, let $\alpha'$ be the section of $\mathbb{S}|\mathcal{S}^{\sim}_{x'}$ over $A'$ deduced from $\alpha$ by applying $u_{\mathbb{S}|\mathcal{S}^{\sim}}$, $\beta'$ the section of $\mathbb{D}|D^{*\sim}_{x'}$ over $\mathrm{Ep}(A')$ deduced from $\beta$ by applying $u_{\mathbb{D}|D^{*\sim}}$, then $\alpha'$ and $\beta'$ are associated by the correlation $h'$.

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*Note marginale.*

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